CS2 AI COACHING SYSTEM

Book-Coach-1B-en

Autore: Renan Augusto Macena

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Ultimate CS2 Coach — Part 1B: The Senses and the Specialist

Topics: RAP Coach Model (7-component architecture: Perception, LTC+Hopfield Memory, Strategy, Pedagogy, Causal Attribution, Positioning, Communication), ChronovisorScanner (multi-scale critical moment detection), GhostEngine (4-tensor inference pipeline), and all external data sources (Demo Parser, HLTV, Steam, FACEIT, TensorFactory, FrameBuffer, FAISS, Round Context).

Author: Renan Augusto Macena

This document is the continuation of **Part 1A** — *The Brain: Neural Architecture and Training*, which documents the neural network models (JEPA, VL-JEPA, AdvancedCoachNN), the 3-level maturity training system, the Coach Introspection Observatory, and the system's architectural foundations (NO-WALLHACK principle, 25-dim contract).

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Part 1A — The Brain: Neural Network Core (JEPA, VL-JEPA, AdvancedCoachNN, SuperpositionLayer, EMA, CoachTrainingManager, TrainingOrchestrator, ModelFactory, NeuralRoleHead, MaturityObservatory)

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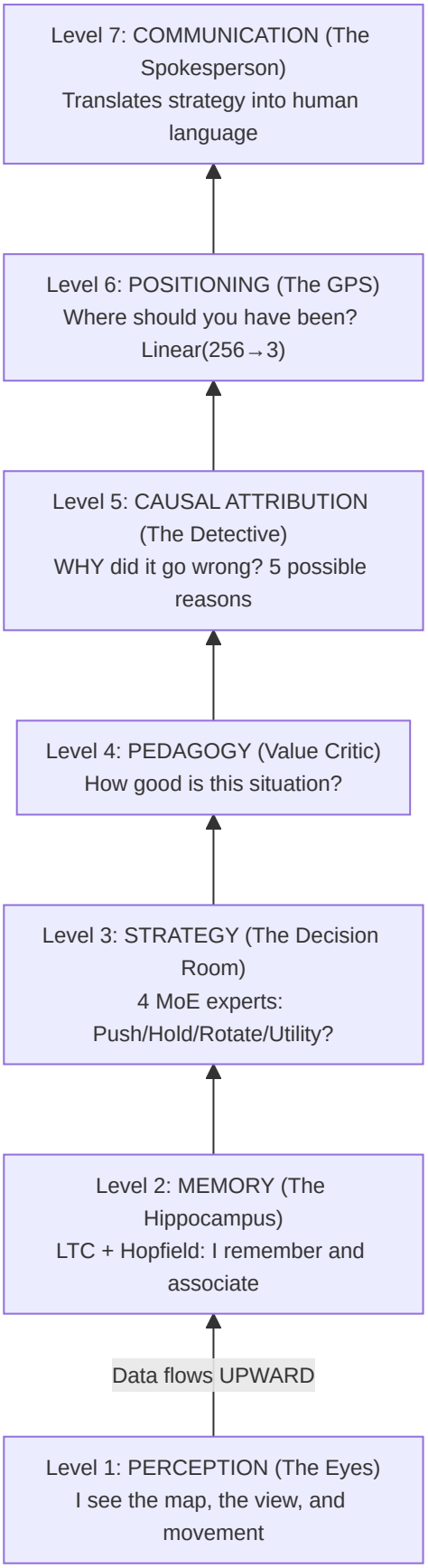
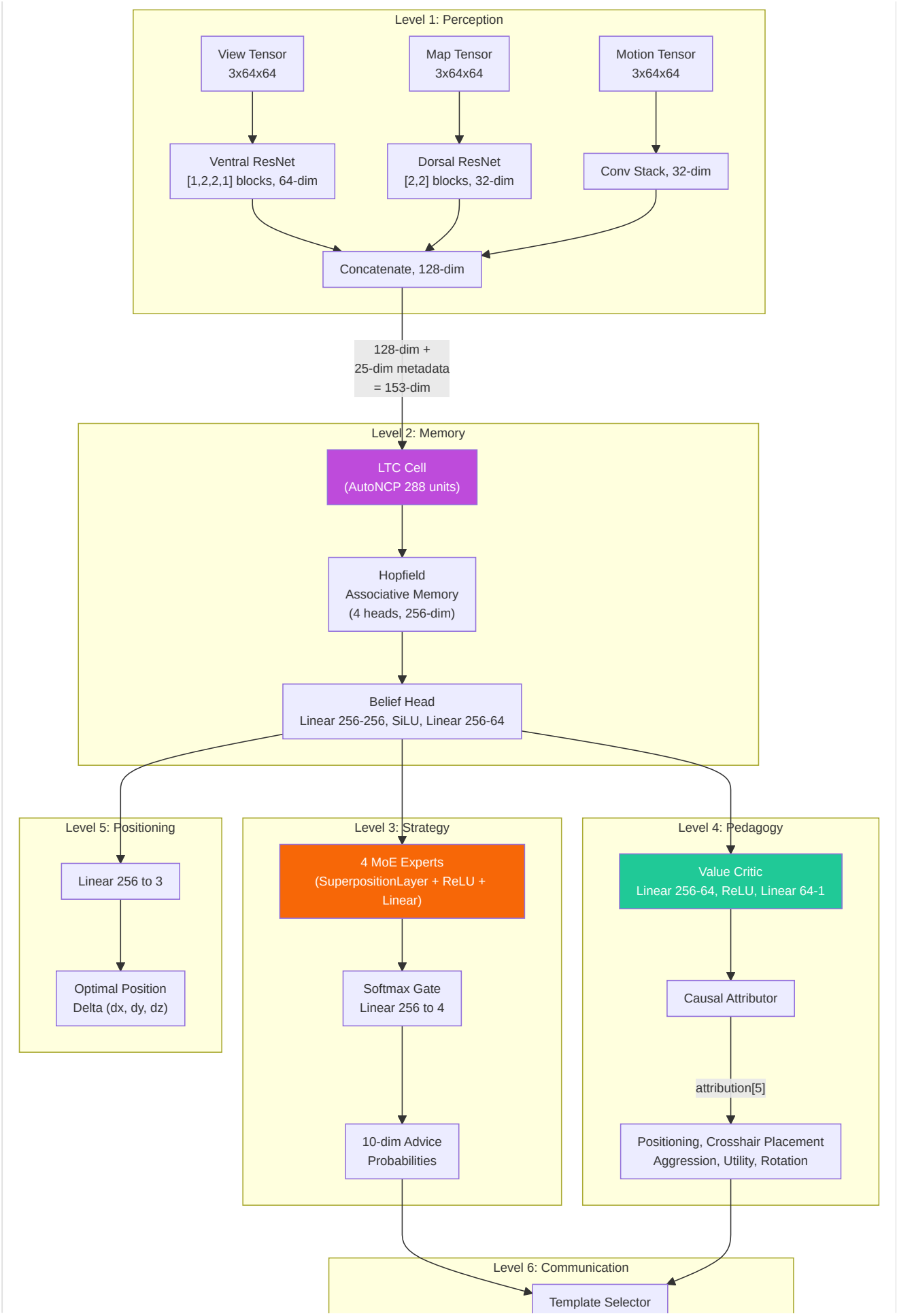
4. Subsystem 2 — RAP Coach Model

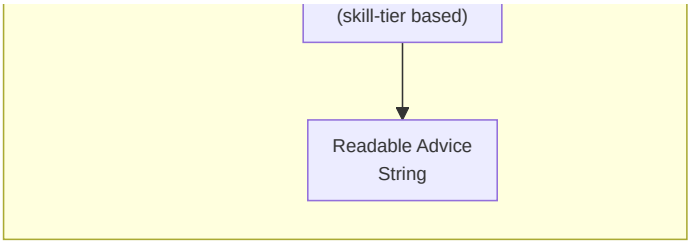
Canonical directory: `backend/nn/experimental/rap_coach/` (the old path `backend/nn/rap_coach/` is a redirection shim) **Files:** `model.py`, `perception.py`, `memory.py`, `strategy.py`, `pedagogy.py`, `communication.py`, `skill_model.py`, `trainer.py`, `chronovisor_scanner.py`

The RAP (Reasoning, Adaptation, Pedagogy) Coach is a **deep architecture with 6 learnable neural components + 1 external communication layer**, specifically designed for CS2 coaching under partial observability conditions (POMDP conditions). The `RAPCoachModel` class contains Perception (`RAPPerception`), Memory (`RAPMemory` with LTC+Hopfield), Strategy (`RAPStrategy`), Pedagogy (`RAPPedagogy` with Value Critic and Skill Adapter), Causal Attribution (`CausalAttributor`) and a Positioning Head (`nn.Linear(256→3)`), all learnable. The Communication layer (`communication.py`) operates externally as a post-processing template selector. The forward pass produces 6 outputs: `advice_probs`, `belief_state`, `value_estimate`, `gate_weights`, `optimal_pos` and `attribution`.

> **Analogy:** The RAP coach is the **most advanced brain** of the system: imagine it as a 7-story building where each floor has a specific task. Floor 1 (Perception) consists of the **eyes**: it observes the map images, the player's view, and movement patterns. Floor 2 (Memory) is the **hippocampus**: it remembers what happened earlier in the round and links it to similar previous rounds via LTC + Hopfield network. Floor 3 (Strategy) is the **decision room**: it decides which advice to give through 4 MoE experts. Floor 4 (Pedagogy) is the **teacher's office**: it estimates the value of the situation with the Value Critic. Floor 5 (Causal Attribution) is the **detective**: it figures out WHY something went wrong, splitting the blame into 5 categories. Floor 6 (Positioning) is the **GPS**: it calculates where the player should have been with an `nn.Linear(256→3)` that predicts `(dx, dy, dz)`. Floor 7 (Communication) is the **spokesperson**: it translates everything into simple readable advice, operating as external post-processing. The "POMDP" part means the coach has to work with **incomplete information**: it cannot see the entire map, just like a player. It is like training a soccer team from the stands when half the field is covered in fog.


Level 7: COMMUNICATION (The Spokesperson)
Translates strategy into human language
Level 6: POSITIONING (The GPS)
Where should you have been?
Linear(256→3)
Level 5: CAUSAL ATTRIBUTION (The Detective)
WHY did it go wrong? 5 possible reasons
Level 4: PEDAGOGY (Value Critic)
How good is this situation?
Level 3: STRATEGY (The Decision Room)
4 MoE experts:
Push/Hold/Rotate/Utility?
Level 2: MEMORY (The Hippocampus)
LTC + Hopfield: I remember and associate
Data flows UPWARD
Level 1: PERCEPTION (The Eyes)
I see the map, the view, and movement


Level 1: Perception
View Tensor
3x64x64
Map Tensor
3x64x64
Motion Tensor
3x64x64
Ventral ResNet
[1,2,2,1] blocks, 64-dim
Dorsal ResNet
[2,2] blocks, 32-dim
Conv Stack, 32-dim
Concatenate, 128-dim
128-dim +
25-dim metadata
= 153-dim
Level 2: Memory
LTC Cell
(AutoNCP 288 units)
Hopfield
Associative Memory
(4 heads, 256-dim)
Belief Head
Linear 256-256, SiLU, Linear 256-64
Level 5: Positioning
Linear 256 to 3
Optimal Position
Delta (dx, dy, dz)
Level 3: Strategy
4 MoE Experts
(SuperpositionLayer + ReLU +
Linear)
Softmax Gate
Linear 256 to 4
10-dim Advice
Probabilities
Level 4: Pedagogy
Value Critic
Linear 256-64, ReLU, Linear 64-1
Causal Attributor
attribution[5]
Positioning, Crosshair Placement
Aggression, Utility, Rotation
Level 6: Communication
Template Selector



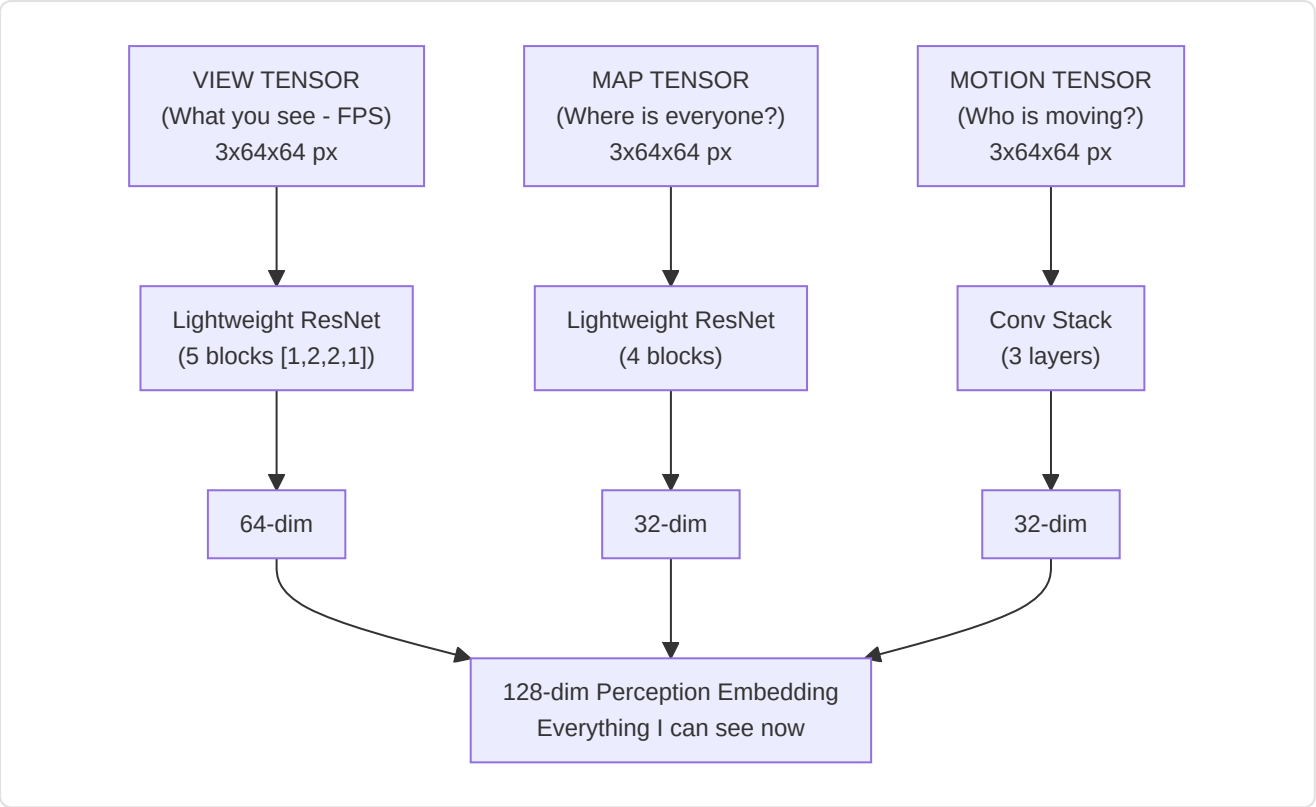
-Perception Layer (`perception.py`)

A **three-stream convolutional** front-end that processes the visual inputs:

Input	Shape	Backbone	Output Dim
View tensor	`3×64×64`	Ventral stream ResNet: [1,2,2,1] blocks, 3→64 channels	**64-dim**
Map tensor	`3×64×64`	Dorsal stream ResNet: [2,2] blocks, 3→32 channels	**32-dim**
Motion tensor	`3×64×64`	Conv(3→16→32) + MaxPool + AdaptiveAvgPool	**32-dim**

The three feature vectors are concatenated into a single **128-dimensional perception embedding** (64 + 32 + 32).

> **Analogy:** The Perception Layer is like the coach's **three different pairs of glasses**. The first pair (view tensor / ventral stream) shows **what the player sees** — their first-person perspective, processed through a lightweight 5-block ResNet (configuration `[1,2,2,1]`, calibrated for 64×64 inputs) that extracts 64 important features from the image. The second pair (map tensor / dorsal stream) shows the **radar/aerial minimap** — where everyone is — processed through a simpler 3-block network into 32 features. The third pair (motion tensor) shows **who is moving and at what speed** — like motion blur in a photo — processed into another 32 features. Then all three views are **glued together** into a single 128-number summary: "Here is everything I can see right now". This process draws inspiration from how the human brain processes vision: the ventral stream recognizes "what" things are, while the dorsal stream tracks "where" things are located.



The ResNet blocks use **identity shortcuts** with learnable downsample (Conv1×1 + BatchNorm) when stride ≠ 1 or the channel count changes. **24 convolution layers** across all three streams:

Stream	Block configuration	Blocks	Conv/Block	Shortcut convs	Total
View (Ventral)	`[1,2,2,1]` → 1 + 5 = 6 blocks	6	2	1 (first block)	**13**
Map (Dorsal)	`[2,2]` → 1 + 3 = 4 blocks	4	2	1 (first block)	**9**
Motion	Conv stack (2 layers)	—	—	—	**2**
Total					**24**

How `_make_resnet_stack` works: It creates 1 initial block with `stride=2` (for spatial downsampling), then `sum(num_blocks) - 1` additional blocks with `stride=1`. Each `ResNetBlock` has 2 Conv2d layers (3×3 kernel). The first block also receives a Conv1×1 shortcut because the input channels (3) differ from the output channels (64 or 32).

Note on architectural choice (F3-29): The original configuration `[3,4,6,3]` (15 blocks, 33 conv in the ventral stream) was designed for 224×224 inputs (ImageNet's standard size). For 64×64 inputs as used in this project, the feature maps would collapse spatially after the first stride-2 block, making subsequent blocks redundant. The `[1,2,2,1]` configuration (5 effective blocks) is specifically calibrated for the 64×64 training resolution, with `AdaptiveAvgPool2d` handling any residual spatial resolution. Any previous checkpoints are automatically detected as `_stale_checkpoint` by `load_nn()`.

> **Analogy:** Identity shortcuts are like the **elevators of a building**: they allow information to skip floors and pass directly from the initial levels to the later ones. Without them, information would have to climb many flights of stairs, and by the time it reached the top, the original signal would be so faded that the network could not learn. Shortcuts ensure that even in a deep network, gradients (the learning signals) can flow efficiently. This is the same trick that made modern deep learning possible, invented by Kaiming He in 2015. Choosing a more compact network (`[1,2,2,1]` instead of `[3,4,6,3]`) is like choosing a 6-story building instead of a 16-story one when the available lot (64×64 pixels) is small: fewer floors mean fewer elevators needed, but transport remains equally efficient.

-Memory Layer (`memory.py`) — LTC + Hopfield

This part addresses the fundamental challenge that the CS2 coach is a **Partially Observable Markov Decision Process** (POMDP).

> **Analogy:** POMDP is a fancy way of saying **"you cannot see everything".** In CS2, you do not know where all the enemies are: you only see what is in front of you. It is like playing chess with a blanket over half the board. The Memory Layer's task is to **remember and guess**: it keeps track of what happened earlier in the round and uses that memory to fill in the blanks about what it cannot see. It has two special tools for this: an LTC network (short-term memory that adapts to the speed of the game) and a Hopfield network (long-term pattern search that says "this situation reminds me of something I have seen before").

Liquid Time-Constant (LTC) network with AutoNCP wiring:

- Input: 153 dim (128 perception + 25 metadata)
- NCP units: **512** (`hidden_dim * 2` = 256 × 2) — a 2:1 ratio that guarantees enough inter-neurons for sparse AutoNCP wiring
- Output: 256-dim hidden state
- Uses the `ncps` library with sparse connectivity patterns, similar to those of the brain
- Adapts temporal resolution to the pace of the game (slow setups vs. fast firefights)
- Deterministic seeding (NN-MEM-02): numpy + torch RNG seeded at 42 during AutoNCP wiring creation, with restoration of the original RNG state after initialization — guarantees checkpoint portability across different runs

> **Analogy:** The LTC network is like a **living, breathing brain**: unlike normal neural networks that process time at fixed intervals (like a clock ticking every second), the LTC adapts its speed to what is happening. During a slow setup (players walking silently), processing happens in slow motion. During a fast firefight, it speeds up, like the heart rate accelerating when excited. "AutoNCP wiring" makes the connections between neurons sparse and structured as in a real brain: not everything connects to everything else. This is more efficient and biologically more realistic.

Hopfield associative memory:

- Input/Output: 256-dim
- Heads: 4
- Uses `hflayers.Hopfield` as **content-addressable memory** for prototype round retrieval

> **Analogy:** Hopfield memory is like a **photo album of famous plays**. During training, it memorizes "prototype rounds" — classic patterns like "a perfect B site retake on Inferno" or "a failed smoke rush on Dust2". When a new moment of play arrives, the Hopfield network asks: "Does this remind me of any photo in my album?" If it finds a match, it retrieves the associated memory, like a police detective flipping through mugshots and saying: "I have seen this face before!" It has 4 "heads" (attention heads) so it can search for 4 different types of patterns simultaneously.

Hopfield activation delay (NN-MEM-01 + RAP-M-04):

The Hopfield network **does not activate immediately** during training. The memorized patterns start from random initialization (`torch.randn * 0.02`) and the attention would be nearly uniform across all slots, adding noise rather than signal. For this reason:

- `_training_forward_count` counts the forward passes during training
- `_hopfield_trained` (boolean flag) remains `False` until ≥2 training forward passes
- Before activation, the forward pass returns `torch.zeros_like(ltc_out)` instead of the Hopfield output
- After ≥2 forwards (ensuring that at least one backward + optimizer.step has shaped the patterns), Hopfield activates and contributes to the combined_state
- Loading a checkpoint (`load_state_dict`) sets `_hopfield_trained = True` immediately, assuming the model has already been trained

> **Analogy:** It is like a **new employee observing for the first 2 days** before being able to make decisions. The Hopfield's photo album is empty at first — the photos are blurry and random. It would be harmful to consult an album of unreadable photos to make tactical decisions. After 2 training passes, the employee has seen enough examples to have at least some meaningful photos in the album, and from that moment starts to actively contribute.

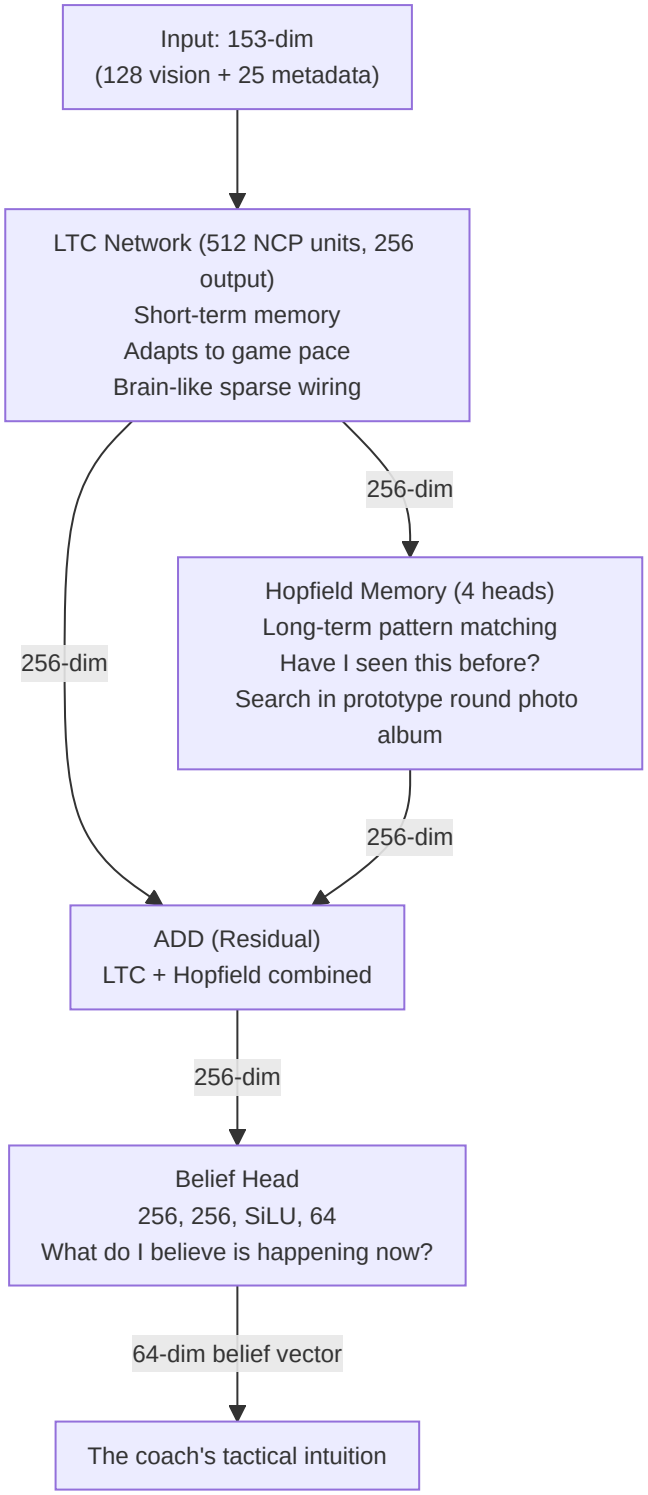
RAPMemoryLite — Pure LSTM fallback:

Lightweight replacement module for `RAPMemory` , used when the `ncps` / `hflayers` dependencies are not available or when a more portable model is desired:

- Standard PyTorch LSTM: `nn.LSTM(153, 256, batch_first=True)`
- Same I/O contract: Input `[B, T, 153]` → Output `(combined_state [B, T, 256], belief [B, T, 64], hidden)`
- Same belief head: `Linear(256→256) → SiLU → Linear(256→64)`
- No RNG seeding needed (no AutoNCP)

- No Hopfield training delay (no memorized patterns)
- Instantiated via `ModelFactory.TYPE_RAP_LITE` ("rap-lite") with `use_lite_memory=True`

> **Analogy:** RAPMemoryLite is like a **backup generator** that runs on simpler fuel. It does not have the "liquid brain" (LTC) that adapts to the pace of the game, nor the photo album (Hopfield) that remembers famous plays. Instead, it uses a traditional LSTM memory — less sophisticated, but reliable and working anywhere without special components. It is Plan B for when the experimental lab is not accessible.



Residual combination: `combined_state = ltc_out + hopfield_out`

> **Analogy:** The residual combination is like **asking two consultants and adding up their opinions**. The LTC says "based on what just happened, I think X". The Hopfield says "based on my memory of similar situations, I think Y". Instead of choosing one, the system adds both opinions: this way, both recent events and historical patterns contribute to the final understanding.

Belief head: `Linear(256→256) → SiLU → Linear(256→64)` — produces a 64-dimensional belief vector that encodes the coach's latent tactical understanding.

Forward pass:

```
ltc_out, hidden = self.ltc(x, hidden) # x: [B, seq, 153] → [B, seq, 256]
mem_out = self.hopfield(ltc_out) # [B, seq, 256]
combined_state = ltc_out + mem_out # Residual
belief = self.belief_head(combined_state) # [B, seq, 64]
return combined_state, belief, hidden
```

-Strategy Layer (`strategy.py`) — Superposition + MoE

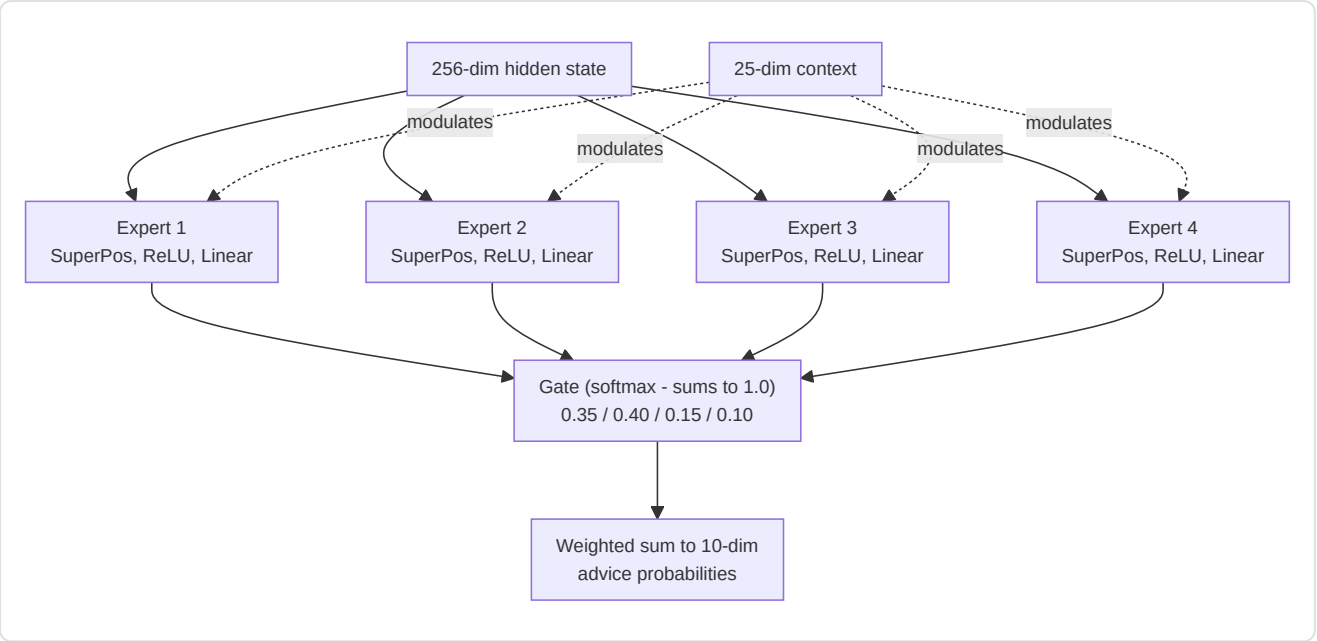
Implements **SuperpositionLayer** combined with a context-conditioned mixture of experts:

> **Analogy:** The Strategy Layer is like a **war room with 4 specialized generals**, each an expert in a different type of situation. One general is good at aggressive pushes, another at defensive holds, another at utility plays, and another still at rotations. A "gatekeeper" (the softmax "gate") listens to the current situation and decides how much to trust each general: "We are in an eco round on Dust2? General 2 (defensive specialist) gets 60% of the power, General 4 (utility) gets 30%, and the others split the rest". The **Superposition Layer** is the secret ingredient: it allows each general to adapt their thinking based on the current game context (map, economy, faction) using an intelligent gating mechanism.

SuperpositionLayers (`layers/superposition.py`): context-dependent gating where `output = F.linear(x, weight, bias) * sigmoid(context_gate(context))` . A sigmoid gate vector conditioned on the **25-dim** context (full METADATA_DIM) selectively masks the expert outputs. The L1 sparsity loss (`context_gate_l1_weight = 1e-4`) encourages sparse and interpretable gating. Observable: gate statistics (mean, std, sparsity, active_ratio) can be tracked.

> **Note:** `RAPStrategy.__init__` uses `context_dim=25` (METADATA_DIM). The gate network is `Linear(hidden_dim=256, num_experts=4) → Softmax(dim=-1)` .

> **Analogy:** The superposition layer is like a **dimmer switch for each neuron**. Instead of having each neuron always fully on, a context-dependent gate (controlled by the 25 metadata features) can dim or boost the brightness of each one. If the context says "this is an eco round", some neurons are dimmed (not relevant for eco rounds), while others are boosted. The L1 sparsity loss is like telling the system: "Try to use as few neurons as possible — the simpler your explanation, the better". This makes the model more interpretable: you can actually see which gates activate in which situations.



4 Expert Modules: Each expert is a `ModuleDict`: `SuperpositionLayer(256→128, context_dim=25) → ReLU → Linear(128→10)`.

Gate Network: `Linear(256→4) → Softmax`.

Output: 10-dimensional advice probability distribution and 4-dimensional gate weights vector.

-Pedagogy Layer (`pedagogy.py`) — Value + Attribution

Two submodules:

1. **Value Critic:** `Linear(256→64) → ReLU → Linear(64→1)`. Estimates V(s) for temporal-difference learning. **Skill Adapter:** `Linear(10 skill_buckets → 256)` enables skill-conditioned value estimates.

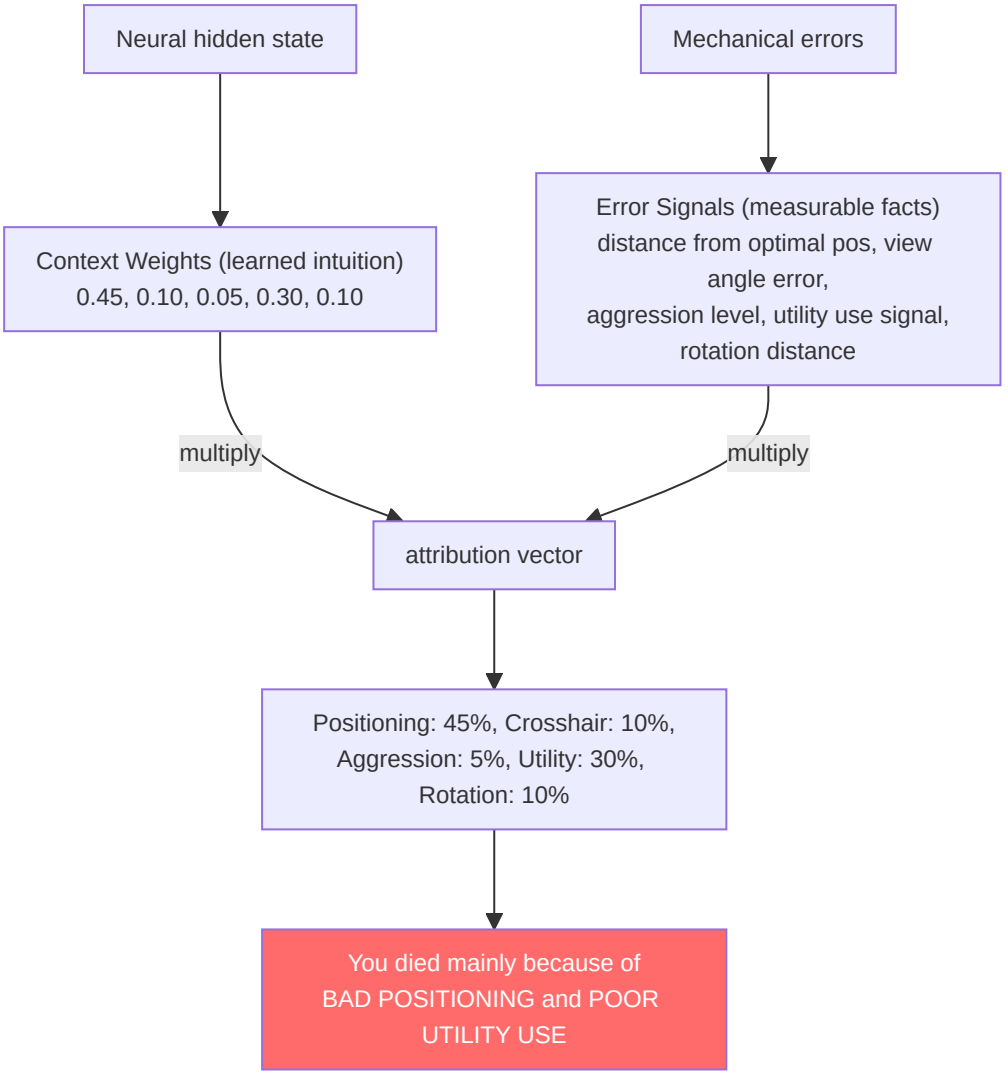
> **Analogy:** The Value Critic is like a **sports commentator** who, at any moment during a match, can say "Right now, this team has a 72% advantage". It estimates V(s) — the "value" of the current state of the match. The **Skill Adapter** adapts this estimate based on the player's skill level: a beginner in the same position as a professional faces very different odds, so the value prediction should reflect this.

1. **CausalAttributor:** Produces a 5-dimensional attribution vector that maps training concepts:

Index	Concept	Mechanical signal
0	**Positioning**	norm(position_delta)
1	**Crosshair placement**	norm(view_delta)
2	**Aggression**	0.5 × position_delta
3	**Utility**	`sigmoid(hidden.mean())` — **learned and context-dependent** signal: produces high activation when the network detects situations where utility use was relevant, low when tactical context makes utility secondary. It is not a static placeholder, but a non-linear function of the hidden state that adapts during training
4	**Rotation**	0.8 × position_delta

Fusion: `attribution = context_weights × mechanical_errors` where context_weights derives from `Linear(256→32) → ReLU → Linear(32→5) → Sigmoid`.

> **Analogy:** The causal attributor is how the coach answers the question **"WHY did it go wrong?"** Instead of just saying "you died", it splits the blame into 5 categories, like a school report card with 5 subjects. "You died because: 45% poor positioning, 30% inadequate utility use, 15% poor crosshair placement, 5% too aggressive, 5% poor rotation." It does this by combining two signals: (1) what the neural network's hidden state considers important (context_weights, the brain's intuition) and (2) measurable mechanical errors (how far from the optimal position, how wrong the viewing angle was). Multiplying them together yields a blame attribution based on both data and intuition.

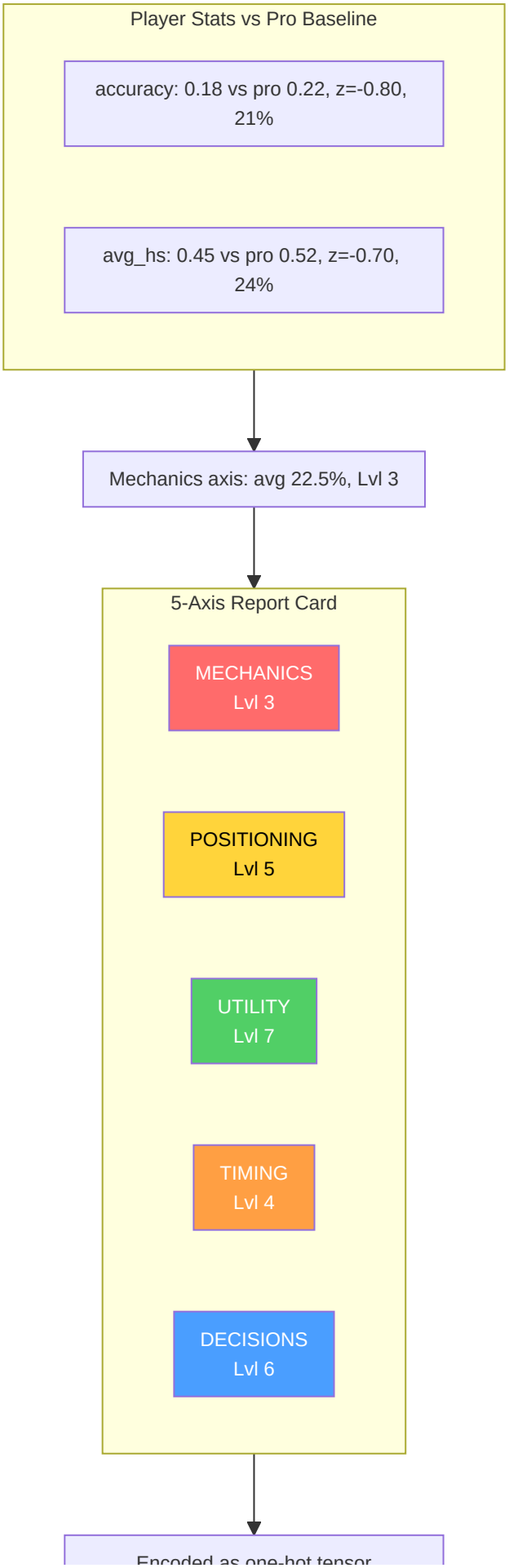


-Latent Skill Model (`skill_model.py`)

Decomposes raw statistics into 5 skill axes using statistical normalization against professional baselines:

Skill Axis	Input statistics	Normalization
Mechanics	Accuracy, avg_hs	Z-score (μ=pro_mean, σ=pro_std)
Positioning	Survival_rating, kast_rating	Z-score
Utility	Utility_blind_time, Utility_enemies_flashed	Z-score
Timing	Opening_duel_win_pct, Positional_aggression_score	Z-score
Decision	Clutch_win_pct, Impact_rating	Z-score

> **Analogy:** The skill model creates a **5-subject report card** for each player. Each subject (Mechanics, Positioning, Utility, Timing, Decision) is graded by comparing the player to professionals. The Z-score is like asking: "How far above or below the class average is this student?". A Z-score of 0 means "exactly average among professionals". A Z-score of -2 means "well below average — needs hard work". A Z-score of +1 means "above average — doing well". The system then converts the Z-scores into percentiles (the percentage of professionals you are better than) and maps them to a curriculum level from 1 to 10, like school grades. A level 1 student receives training suited for beginners; a level 10 student receives advanced tactical analysis.


Player Stats vs Pro Baseline
accuracy: 0.18 vs pro 0.22, z=-0.80, 21%
avg_hs: 0.45 vs pro 0.52, z=-0.70, 24%
Mechanics axis: avg 22.5%, Lvl 3
5-Axis Report Card
MECHANICS Lvl 3
POSITIONING Lvl 5
UTILITY Lvl 7
TIMING Lvl 4
DECISIONS Lvl 6
Encoded as one-hot tensor

Encoded as one-hot tensor
Fed to Pedagogy Layer's Skill Adapter

The Z-scores are converted to percentiles via the **logistic approximation** `1/(1+exp(-1.702z))` (fast CDF approximation), then the mean percentile is mapped to a **curriculum level** (1–10) via `int(avg_skill * 9) + 1`, clamped to [1, 10]. The level is encoded as a one-hot tensor (10-dim) via `SkillLatentModel.get_skill_tensor()` for the Pedagogy Layer's Skill Adapter.

-RAP Trainer (`trainer.py`)

Orchestrates the training loop with a **composite loss function**:

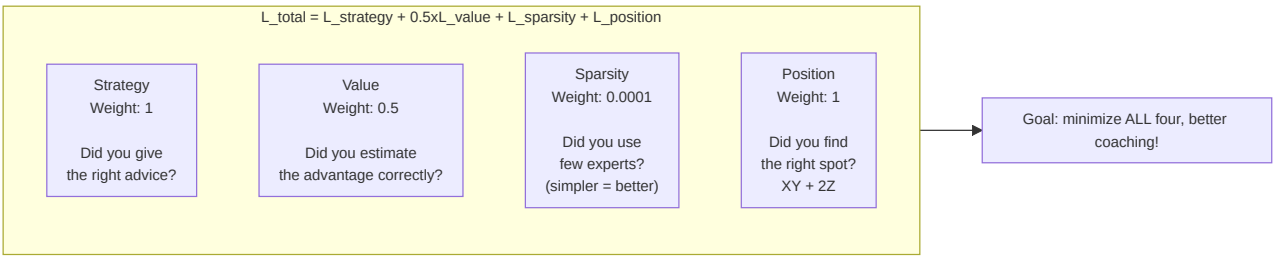
```
L_total = L_strategy + 0.5 × L_value + L_sparsity + L_position
```

> **Analogy:** The total loss is like a **report card with 4 grades**, each of which measures a different aspect of the model's performance. The model tries to make ALL four grades as low as possible (in machine learning, lower loss = better performance). The weights (1.0, 0.5, 1e-4, 1.0) indicate the importance of each subject: Strategy and Position are full-score subjects, Value is half credit, and Sparsity is extra credit. The model cannot just pass one subject and fail the others: it must balance all four.

Loss term	Formula	Weight	Purpose
`L_strategy`	`MSELoss(advice_probs, target_strat)`	1.0	Correct tactical recommendation
`L_value`	`MSELoss(V(s), true_advantage)`	0.5	Accurate advantage estimation
`L_sparsity`	`model.compute_sparsity_loss(gate_weights)` — L1 on gate weights (explicit parameter, thread-safe)	1e-4	Expert specialization
`L_position`	`MSE(pred_xy, true_xy) + 2.0 × MSE(pred_z, true_z)`	1.0	Optimal positioning, **strict penalty on the Z-axis**

> **Note:** The 2× multiplier on the Z-axis exists because vertical positioning errors (e.g., a wrong floor on Nuke/Vertigo) are tactically catastrophic: they represent wrong-floor errors that no horizontal correction can fix.

> **Analogy:** The Z-axis penalty is like a **fire alarm for wrong-floor errors**. On CS2 maps like Nuke (which has two floors) or Vertigo (a skyscraper), telling a player to go to the wrong floor is a disaster: it is like telling someone to go to the kitchen when you meant the attic. Being slightly off horizontally (X/Y) is like being a few steps left or right — not great, but fixable. Being on the wrong floor (Z) is like being in a completely different room. That is why vertical errors are punished 2× harder during training: the model quickly learns to "NEVER suggest the wrong floor".



Output per training step: `{loss, sparsity_ratio, loss_pos, z_error}`.

-RAPCoachModel Forward Pass Summary

Signature: `forward(view_frame, map_frame, motion_diff, metadata, skill_vec=None, hidden_state=None)` — the `hidden_state` parameter (NN-40) allows passing the recurrent memory state from one call to the next, enabling **continuous inference** without cold-start: the GhostEngine can maintain memory between consecutive ticks rather than starting from scratch at every evaluation.

NN-39 Fix — Dual Visual Inputs: The forward pass handles two formats of visual input through an explicit dimensional check:

Input Format	Shape	When used	Behavior
Per-timestep	`[B, T, C, H, W]` (5-dim)	Training with temporal sequences	Each timestep processed individually by the CNN
Static	`[B, C, H, W]` (4-dim)	Real-time inference (GhostEngine)	Single frame expanded across all timesteps

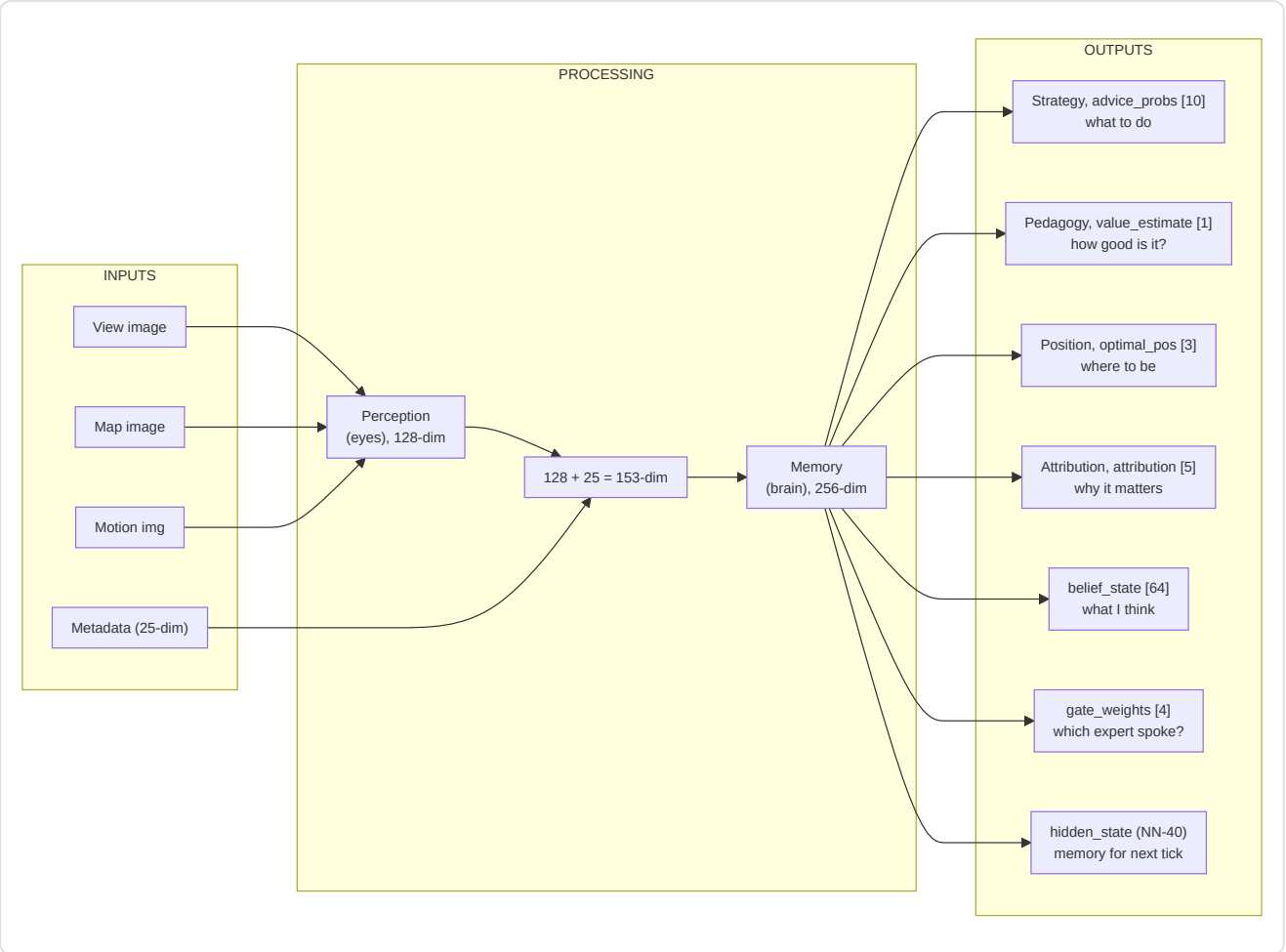
> **NN-39 Analogy:** Imagine showing a film to the coach. In the **per-timestep** format, the coach watches each frame one by one, analyzing them separately and building an understanding that evolves over time — like a referee reviewing an action in slow motion, frame by frame. In the **static** format, the coach sees a single photograph of the situation and assumes the scene has remained unchanged for the entire duration — as when analyzing a position from a screenshot. The NN-39 fix ensures that both situations produce the same output format (`[B, T, 128]`), so the rest of the brain (memory, strategy, pedagogy) works identically in both cases.

```
def forward(view_frame, map_frame, motion_diff, metadata, skill_vec=None):
    batch_size, seq_len, _ = metadata.shape

    # NN-39 fix: supports per-timestep visual input [B,T,C,H,W] and static [B,C,H,W]
    if view_frame.dim() == 5:
        # Per-timestep — process each timestep through the CNN separately
        z_frames = []
        for t in range(view_frame.shape[1]):
            z_t = self.perception(view_frame[:, t], map_frame[:, t], motion_diff[:, t])
            z_frames.append(z_t)
        z_spatial_seq = torch.stack(z_frames, dim=1)      # [B, T, 128]
    else:
        # Static — single frame expanded across all timesteps
        z_spatial = self.perception(view_frame, map_frame, motion_diff)  # [B, 128]
        z_spatial_seq = z_spatial.unsqueeze(1).expand(-1, seq_len, -1)   # [B, T, 128]

    lstm_in = cat([z_spatial_seq, metadata], dim=2)        # [B, seq, 153]
    hidden_seq, belief, new_hidden = self.memory(lstm_in, hidden=hidden_state)  # [B, seq, 256]
    last_hidden = hidden_seq[:, -1, :]
    prediction, gate_weights = self.strategy(last_hidden, context)  # [B, 10], [B, 4]
    value_v = self.pedagogy(last_hidden, skill_vec)        # [B, 1]
    optimal_pos = self.position_head(last_hidden)          # [B, 3]
    attribution = self.attributor.diagnose(last_hidden, optimal_pos) # [B, 5]
    return {
        "advice_probs": prediction,      # [B, 10]
        "belief_state": belief,          # [B, seq, 64]
        "value_estimate": value_v,       # [B, 1]
        "gate_weights": gate_weights,    # [B, 4]
        "optimal_pos": optimal_pos,      # [B, 3]
        "attribution": attribution,      # [B, 5]
        "hidden_state": new_hidden,      # NN-40: recurrent state for continuous inference
    }
```

Analogy: This is the **complete recipe** of how the RAP Coach thinks, step by step: (1) **Eyes** — the Perception layer examines the view, map, and motion images and creates a 128-number summary of what it sees. The NN-39 fix allows two modes: if it receives a film (5-dim), it processes each frame separately; if it receives a photo (4-dim), it replicates it across all timesteps. (2) This visual summary is combined with 25 metadata numbers (health, position, economy, etc.) to form a 153-number description. (3) **Memory** — the LTC + Hopfield memory processes the description over time, producing a 256-number hidden state and a 64-number belief vector ("what I think is happening"). (4) **Strategy** — 4 experts examine the hidden state and produce 10 advice probabilities ("40% chance you should push, 30% hold, etc."). (5) **Teacher** — the pedagogy layer estimates "how good is this situation?" (value). (6) **GPS** — the position head predicts where you should move (3D coordinates). (7) **Blame** — the attributor figures out why things went wrong (5 categories). All **7** outputs are returned together as a dictionary: the complete training analysis for a moment of play. The seventh output, `hidden_state` (NN-40), is the recurrent memory state — it allows the GhostEngine to maintain "memory" between consecutive ticks, like a coach who does not forget what happened 5 seconds ago when evaluating the current position.



-ChronovisorScanner (`chronovisor_scanner.py`)

A **multi-scale signal processing module** that identifies critical moments in matches by analyzing temporal advantage deltas at **3 resolution levels** (micro, standard, macro):

> **Analogy:** The Chronovisor is like a **highlight detector with 3 magnifying lenses**. The **micro** lens (sub-second) captures instantaneous decisions in firefights — like a referee reviewing an action in slow motion. The **standard** lens (engagement level) spots critical moments such as decisive plays or fatal mistakes — like the main match replay. The **macro** lens (strategic) detects strategy shifts developing over 5-10 seconds — like the commentator's tactical analysis. It works by monitoring the team advantage over time (like a stock price chart) and looking for sudden spikes or crashes at each scale. Instead of watching the entire 45-minute match, the player can jump directly to the most significant critical moments.

Multi-Scale Configuration (`ANALYSIS_SCALES`):

Scale	Window (ticks)	Lag	Threshold	Description
Micro	64	16	0.10	Sub-second engagement decisions
Standard	192	64	0.15	Engagement-level critical moments
Macro	640	128	0.20	Strategic shift detection (5-10 seconds)

> **Multi-scale analogy:** The three scales are like **three different zoom levels on Google Maps**: the micro scale is the street level (you can see every detail of an intersection), the standard scale is the neighborhood level (you see the overall structure of the area), the macro scale is the city level (you see how neighborhoods connect to one another). A player may have a bad micro-decision (too-slow peek) that does not appear in the larger scales, or a macro strategic change (late rotation) that is not visible in micro-analysis. By using all three simultaneously, the coach captures both instantaneous errors and wrong strategic choices.

Detection pipeline (for each scale):

1. Uses the trained RAP model to predict V(s) for each tick window.
2. Computes deltas using the lag configured for the scale: `deltas = values[LAG:] - values[:-LAG]`.
3. Detects **spikes** where `|delta| > threshold` (variable per scale: 0.10/0.15/0.20).
4. Searches for the peak within the configured window, maintaining sign consistency.
5. **Non-maximum suppression** prevents duplicate detections.
6. Classifies each peak as **"play"** (positive gradient, advantage gained) or **"error"** (negative, advantage lost).
7. Returns instances of the `CriticalMoment` dataclass with `(match_id, start_tick, peak_tick, end_tick, severity [0-1], type, description, scale)`.

Tick safety limit (F3-21): `_MAX_TICKS_PER_SCAN = 50,000` — matches with more than 50K ticks (possible with extended overtime or very long matches) are **truncated** with a warning (NN-CV-02) rather than saturating RAM. The system fetches `_MAX_TICKS_PER_SCAN + 1` ticks to detect truncation and warns that critical moments in the late match phase may be lost.

Cross-scale deduplication: When the same moment is detected at different scales (e.g., a critical peek visible in both micro and standard scales), deduplication prioritizes **micro > standard > macro** (the finer scale wins). `MIN_GAP_TICKS = 64` (~1 second) defines the minimum distance between two moments: if two spikes are closer than 64 ticks, they are considered the same event and only the finer-scale one is kept.

Severity labels: Severity (0-1) is automatically classified for the `MatchVisualizer`:

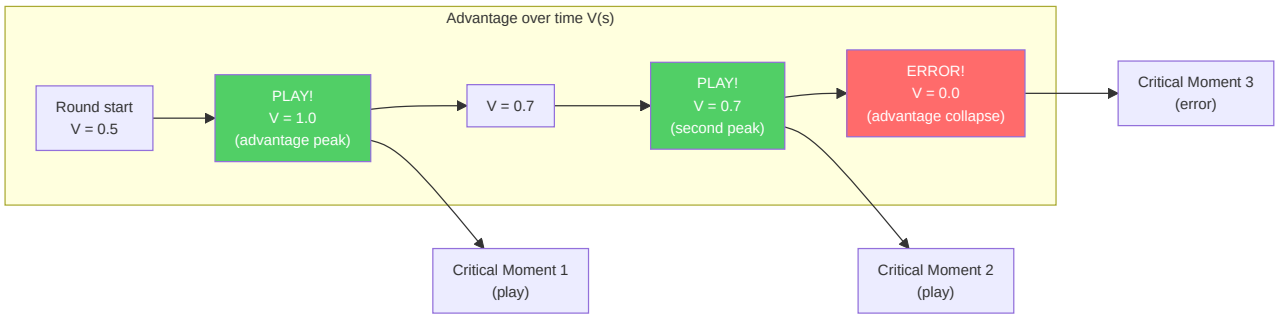
- `severity > 0.3` → **"critical"** (game-changing moment)
- `severity > 0.15` → **"significant"** (relevant moment)
- otherwise → **"notable"** (noteworthy moment)

`ScanResult` **dataclass:** Structured return type that distinguishes success from failure:

Field	Type	Description
`critical_moments`	`List[CriticalMoment]`	Detected critical moments
`success`	`bool`	True if the scan completed (even with 0 moments)
`error_message`	`Optional[str]`	Error detail if `success=False`
`model_loaded`	`bool`	Whether the RAP model was available
`ticks_analyzed`	`int`	Number of ticks actually analyzed

Utility properties: `is_empty_success` (successful scan but no critical moments found), `is_failure` (failed scan — model not loaded, DB error, etc.).

> **Pipeline analogy:** Here is the step-by-step procedure: (1) The RAP model observes each moment and assigns an "advantage score" (like a heart rate monitor). (2) For each of the 3 scales, it compares each moment with what happened N ticks before (16, 64, or 128 ticks depending on scale) — "did things get better or worse?" (3) If the change exceeds the scale's threshold, it is a significant event — like a heart rate spike. (4) It zooms into the window around the peak to find the exact peak moment. (5) It filters out duplicate detections — if two peaks are too close together, it only keeps the larger one. (6) It labels each peak: "play" (you did something excellent) or "error" (you made a mistake). (7) It packages everything into an orderly evaluation sheet for each critical moment, with severity scores from 0 (minor) to 1 (game-changing) and the detection scale (micro/standard/macro).



-GhostEngine (`inference/ghost_engine.py`)

Real-time inference for the "Ghost" — overlay of the player's optimal position. The GhostEngine represents the **endpoint** of the entire neural chain: it is where the RAP Coach Model produces outputs visible to the user in the form of a "ghost player" on the tactical map.

> **Analogy:** The Ghost Engine is like a **"better you" hologram** on the screen. At every moment during playback, it asks the RAP Coach: "Given this exact situation, where SHOULD the player be?" The answer is a small position delta (e.g., "5 pixels to the right and 3 pixels up"), which is scaled to real map coordinates. The result is a transparent "ghost" player displayed on the tactical map, showing the optimal position. If the ghost is far from where you actually were, you know you are in a bad position. If it is close, you positioned well.

4-Tensor Inference Pipeline with PlayerKnowledge:

The inference pipeline operates in 5 sequential phases for each playback tick:

Phase 1 — Model Loading (`_load_brain()`)

- Verifies `USE_RAP_MODEL` from configuration (master switch)
- `ModelFactory.get_model(ModelFactory.TYPE_RAP)` — instantiates the RAP model
- `load_nn(checkpoint_name, model)` — loads the weights from the checkpoint on disk
- `model.to(device)` → `model.eval()` — moves to GPU/CPU and activates inference mode
- On failure: `model = None`, `is_trained = False` — disables predictions

Phase 2 — Input Tensor Construction

Tensor	Method	Output Shape	Content
Map	`tensor_factory.generate_map_tensor(ticks, map_name, knowledge)`	`[1, 3, 64, 64]`	Teammates positions, visible enemies, utility + bomb
View	`tensor_factory.generate_view_tensor(ticks, map_name, knowledge)`	`[1, 3, 64, 64]`	90° FOV mask, visible entities, utility zones
Motion	`tensor_factory.generate_motion_tensor(ticks, map_name)`	`[1, 3, 64, 64]`	32-tick trajectory, velocity field, crosshair delta
Metadata	`FeatureExtractor.extract(tick_data, map_name, context)`	`[1, 1, 25]`	Canonical 25-dim vector (health, position, economy, etc.)

The **PlayerKnowledge bridge** (`_build_knowledge_from_game_state()`) filters data according to the NO-WALLHACK principle: only information legitimately available to the player (teammates, visible enemies, last known positions with decay) is encoded in the map and view tensors. If knowledge construction fails, the system degrades to legacy mode (empty tensors).

Phase 2b — POV Mode (R4-04-01):

Mode	Condition	Behavior
POV Mode	`USE_POV_TENSORS=True` + `game_state` provided	Builds `PlayerKnowledge` from the game state → POV tensors with dedicated channel semantics
Legacy Mode	`USE_POV_TENSORS=False` (default)	Standard tensors aligned with training data

> **Warning (R4-04-01):** POV tensors use different channel semantics (Ch0=teammates, Ch1=last-known enemies) compared to standard training data (Ch0=enemies, Ch1=teammates). Using POV tensors with a model trained in legacy mode will produce **unreliable** results. POV mode is only valid if the model was trained with POV data.

Phase 3 — Neural Inference

```
with torch.no_grad():
    out = self.model(view_frame=view_t, map_frame=map_t,
                     motion_diff=motion_t, metadata=meta_t,
                     hidden_state=self._last_hidden)  # NN-40: persistent state
self._last_hidden = out["hidden_state"]  # Keep for next tick
```

`torch.no_grad()` disables gradient computation (inference only, no training). The `hidden_state` parameter (NN-40) allows maintaining the recurrent memory state between consecutive ticks, avoiding cold-start at each evaluation.

Phase 4 — Decoding and Position Scaling

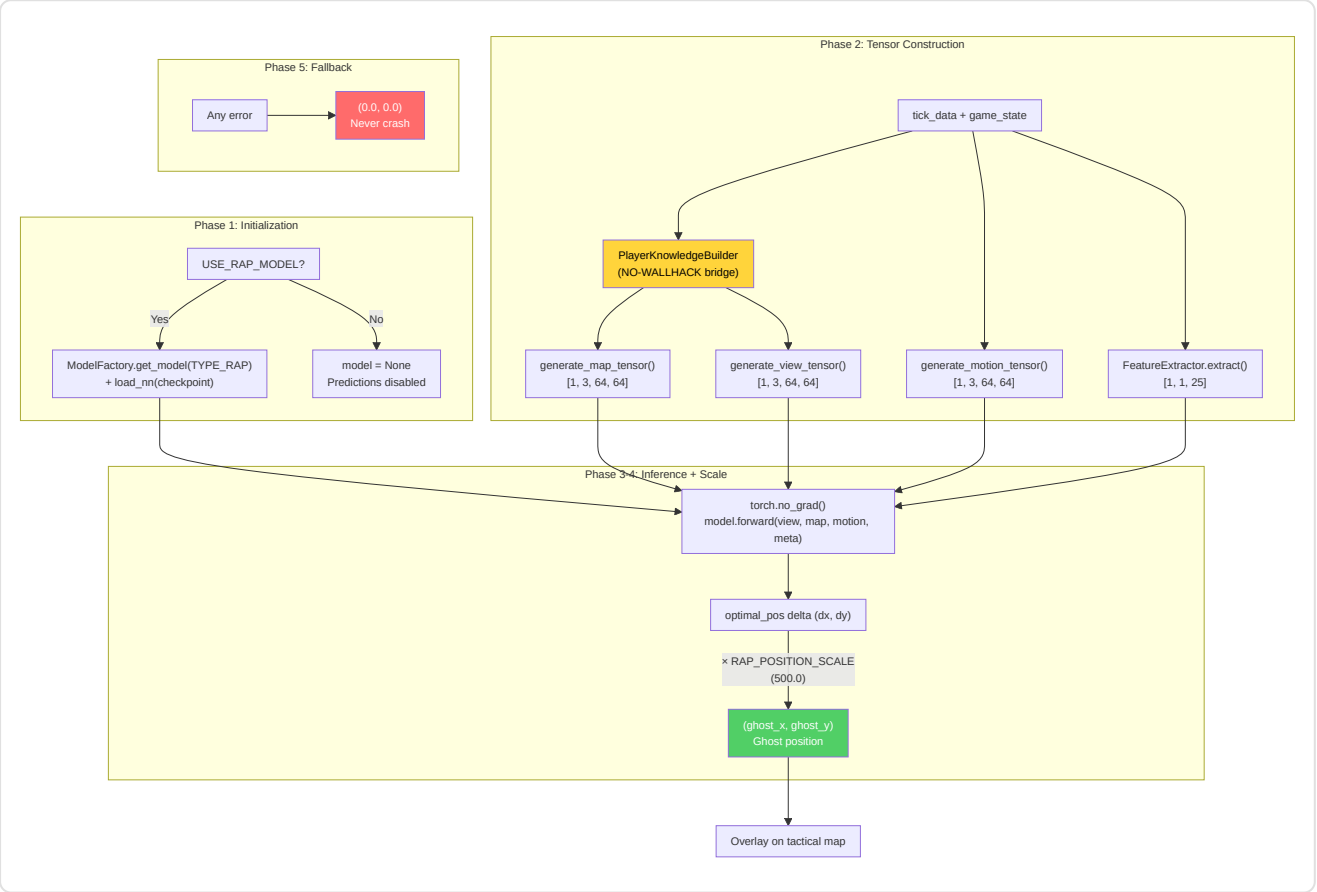
```
optimal_delta = out["optimal_pos"].cpu().numpy()[0]    # [dx, dy, dz]
ghost_x = current_x + (optimal_delta[0] * RAP_POSITION_SCALE)  # × 500.0
ghost_y = current_y + (optimal_delta[1] * RAP_POSITION_SCALE)  # × 500.0
return (ghost_x, ghost_y)
```

The model produces a delta normalized in [-1, 1] that is scaled to world coordinates via `RAP_POSITION_SCALE = 500.0` (from `config.py`). The constant is shared between GhostEngine and overlay to ensure consistency.

Phase 5 — Graceful Fallback (5 modes)

Fallback Mode	Condition	Behavior
Model disabled	`USE_RAP_MODEL=False`	Skip loading, returns `(0.0, 0.0)`
Missing checkpoint	Training not completed	`model = None`, predictions disabled
Missing map name	No spatial context	Returns `(0.0, 0.0)` immediately
PlayerKnowledge error	Knowledge construction failed	Degrades to legacy tensors (all zeros)
Inference error	RuntimeError / CUDA OOM	Logs error, returns `(0.0, 0.0)`

> **Fallback analogy:** The fallback is like a GPS with 5 safety levels: (1) "Offline mode — I have no maps loaded", (2) "I never learned to navigate this area", (3) "I do not even know what city we are in", (4) "I know where we are but cannot see around us — I drive from memory", (5) "Something broke — I just tell you to stay where you are". In every case, the GPS **never sends the car into a wall** — the worst possible response is "stay put" (`(0.0, 0.0)`), which is infinitely better than an application crash.



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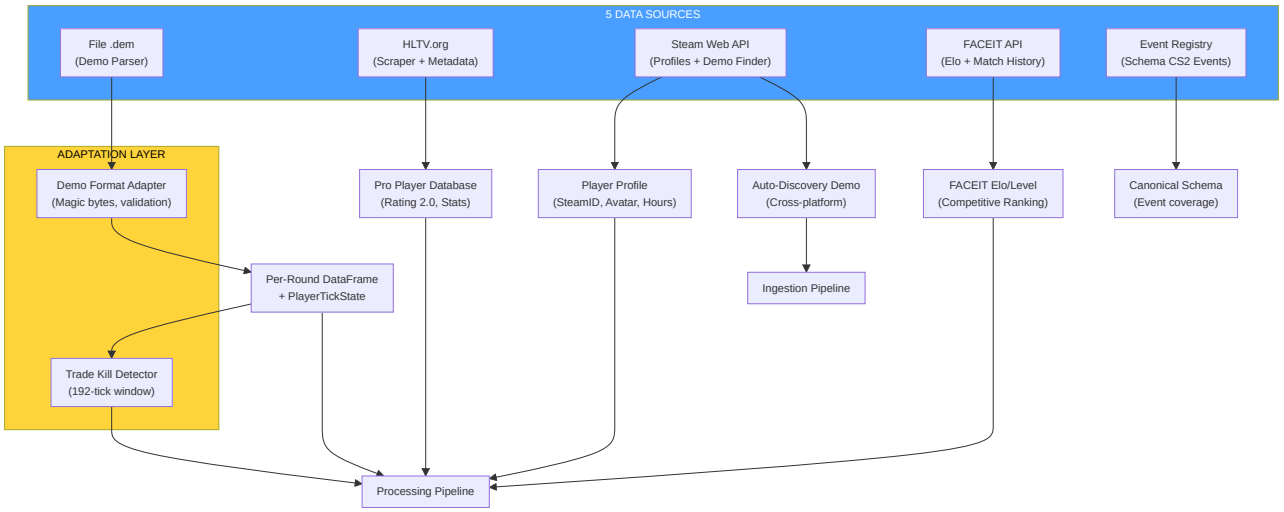
5. Subsystem 1B — Data Sources

Program folder: `backend/data_sources/` **Files:** `demo_parser.py`, `demo_format_adapter.py`, `event_registry.py`, `trade_kill_detector.py`, `hltv_scraper.py`, `hltv_metadata.py`, `steam_api.py`, `steam_demo_finder.py`, `faceit_api.py`, `faceit_integration.py`, `__init__.py`

The Data Sources subsystem is the **entry point for all external data** into the system. It collects information from 5 distinct sources: CS2 demo files, HLTV statistics, Steam profiles, FACEIT data, and game event registry.

> **Analogy:** The Data Sources are like the **5 senses** of the AI coach. The main eye (demo parser) watches match recordings frame by frame. The ear (HLTV scraper) listens to news from the professional world. The touch (Steam API) senses the player's profile and history. The taste (FACEIT) tastes the player's competitive level. The sixth sense (event registry) systematically catalogs every type of event the game can produce. Without these senses, the coach would be blind and deaf — unable to learn anything.



-Demo Parser (`demo_parser.py`)

Robust wrapper around the `demoparser2` library for extracting statistics from CS2 demo files.

HLTV 2.0 Baseline — normalization constants for rating computation:

Constant	Value	Meaning
`RATING_BASELINE_KPR`	0.679	Pro average: kills per round
`RATING_BASELINE_SURVIVAL`	0.317	Pro average: survival rate
`RATING_BASELINE_KAST`	0.70	Pro average: Kill/Assist/Survive/Trade %
`RATING_BASELINE_ADR`	73.3	Pro average: average damage per round
`RATING_BASELINE_ECON`	85.0	Pro average: economic efficiency

`parse_demo(demo_path, target_player=None)`: Main entry point. Validation of file existence, parsing of `round_end` events to count rounds, then complete statistical extraction via `_extract_stats_with_full_fields()`. Returns empty `pd.DataFrame` on any error (fail-safe).

`_extract_stats_with_full_fields(parser, total_rounds, target_player)`: Computes all 25 mandatory aggregate features for the database:

- Base statistics: `avg_kills`, `avg_deaths`, `avg_adr`, `kd_ratio`
- Variance: `kill_std`, `adr_std` (via `_compute_per_round_variance`)

- Advanced statistics: `avg_hs`, `accuracy`, `impact_rounds`, `econ_rating`
- Approximate HLTV 2.0 rating (hand-tuned approximation, not the official formula)

> **Analogy:** The Demo Parser is like an **expert sports commentator** who watches the recording of a match and compiles a detailed report card for each player. It does not just count kills: it calculates damage per round, headshot percentage, economic efficiency, and even how consistent the performances are (standard deviation). If the recording is corrupted or data is missing, the commentator writes "no data available" instead of making up numbers — it is the project's zero-tolerance policy against data fabrication.

-Demo Format Adapter (`demo_format_adapter.py`)

Resilience layer for handling different versions of the CS2 demo format.

Validation constants:

Constant	Value	Description
`DEMO_MAGIC_V2`	`b"PBDEMS2\x00"`	CS2 magic bytes (Source 2 Protobuf)
`DEMO_MAGIC_LEGACY`	`b"HL2DEMO\x00"`	CS:GO legacy magic bytes (not supported)
`MIN_DEMO_SIZE`	10×1024^2 (10 MB)	DS-12: real CS2 demos are 50+ MB, smaller files are certainly corrupted or incomplete
`MAX_DEMO_SIZE`	5×1024^3 (5 GB)	Safety cap

Dataclasses:

- `FormatVersion(name, magic, description, supported)` — specifies a known version of the format
- `ProtoChange(date, description, affected_events, migration_notes)` — record of a known protobuf change

`FORMAT_VERSIONS`: Dictionary with two known formats (`cs2_protobuf` supported, `csgo_legacy` not supported).

`PROTO_CHANGELOG`: Chronological list of known CS2 protobuf format changes (for resilience against future updates).

`DemoFormatAdapter.validate_demo(path)`: 3-phase validation: (1) existence and size within bounds, (2) reading magic bytes for format identification, (3) verification of support for the detected format.

> **Analogy:** The Demo Format Adapter is like a **customs officer at the airport** who inspects each "package" (demo file) before letting it enter the system. It checks: (1) "Is the package the right size?" (not too small = corrupted, not too large = potential bomb), (2) "Does it have the right stamp?" (magic bytes PBDEMS2 = CS2, HL2DEMO = old CS:GO), (3) "Do we accept packages from this country?" (CS2 yes, CS:GO no). If something does not match, the package is rejected with a clear message on why. This prevents corrupted or wrong-format files from entering the pipeline and causing mysterious errors downstream.

-Event Registry (`event_registry.py`)

Canonical registry of **all CS2 game events** derived from SteamDatabase dumps.

`GameEventSpec` dataclass with 7 fields: `name`, `category` (round/combat/utility/economy/movement/meta), `fields` (dict field→type), `priority` (critical/standard/optional), `implemented` (bool), `handler_path` (optional), `notes`.

Registered event categories:

Category	Events	Critical Priority	Implemented
Round	`round_end`, `round_start`, `round_freeze_end`, `round_mvp`, `begin_new_match`	`round_end`	1/5
Combat	`player_death`, `player_hurt`, `player_blind`, etc.	`player_death`	partial
Utility	`flashbang_detonate`, `hegrenade_detonate`, `smokegrenade_expired`, etc.	—	partial
Economy	`item_purchase`, `bomb_planted`, `bomb_defused`, etc.	`bomb_planted/defused`	partial
Movement	`player_footstep`, `player_jump`, etc.	—	no
Meta	`player_connect`, `player_disconnect`, etc.	—	no

Utility functions: `get_implemented_events()` → list of implemented events. `get_coverage_report()` → coverage report by category.

> **Note (F6-33):** The `handler_path` fields are not validated at runtime — if the handler modules are moved, references become silently stale. Add `hasattr/callable` validation at event dispatch if reliability is critical.

> **Analogy:** The Event Registry is like an **encyclopedic catalog of all the signals the game can emit**. Each signal is classified by category (combat, round, utility, economy, movement, meta), priority (critical/standard/optional), and implementation status. It is like a museum catalog: every artwork has a card with title, room, artist, and whether it is currently on display. This allows the team to know exactly which events the system handles and which are missing, planning expansion systematically.

-Trade Kill Detector (`trade_kill_detector.py`)

Identifies **trade kills** — retaliation kills within a temporal window — from the death sequences in the demo.

Constant: `TRADE_WINDOW_TICKS = 192` (3 seconds at 64 ticks/sec, the standard CS2 tickrate).

`TradeKillResult` dataclass:

- `total_kills`, `trade_kills`, `players_traded`, `trade_details`
- Computed properties: `trade_kill_ratio`, `was_traded_ratio`

Algorithm (derived from cstat-main): For each kill K at tick T: look back in time for kills made by the victim. If the victim killed a teammate of K's killer within `TRADE_WINDOW_TICKS`, mark K as a trade kill and the original victim as "was traded". **Same-round constraint:** Kills candidate for the trade must belong to the **same round** (identical `round_num`). Cross-round trades are not counted — this is an important tactical distinction because a trade has strategic meaning only within the same round, where it directly influences the numerical economy of the engagement.

`build_team_roster(parser)`**:** Builds `player_name → team_num` mapping from the initial match ticks (uses the 10th percentile of ticks for assignment stability).

`get_round_boundaries(parser)`**:** Extracts the tick boundaries between rounds from the `round_end` event.

> **Analogy:** The Trade Kill Detector is like a **sports replay analyst** who reviews each elimination and asks: "Did anyone avenge this player within 3 seconds?" If yes, the death was "traded" — meaning the team reacted quickly. A high trade kill ratio indicates good team coordination; a low ratio indicates isolated players dying without support. This metric is one of the most important indicators in professional CS2 for evaluating positional discipline and team communication.

-Steam API (`steam_api.py`)

Client for the Steam Web API with retry and exponential backoff.

Constants: `MAX_RETRIES = 3`, `BACKOFF_DELAYS = [1, 2, 4]` seconds.

`_request_with_retry(url, params, timeout=5)`**:** HTTP GET wrapper with 3 attempts for connection/timeout errors. Does not retry on HTTP 4xx/5xx errors (propagates them to the caller).

Main functions:

- `resolve_vanity_url(vanity_url, api_key)` → resolves a custom Steam URL to a 64-bit SteamID
- `fetch_steam_profile(steam_id, api_key)` → retrieves player profile (name, avatar, playtime hours). Auto-resolves vanity URL if the input is not numeric

-Steam Demo Finder (`steam_demo_finder.py`)

Auto-discovery of CS2 demos from the local Steam installation.

`SteamDemoFinder` class with 3-level detection strategy:

Priority	Method	Platform
1	Windows Registry (`winreg`)	Windows
2	Common paths (dynamically generated for each drive)	Windows/Linux/macOS
3	Environment variables	All

Dynamic drive detection (Windows): Uses `windll.kernel32.GetLogicalDrives()` to enumerate all available drives, then searches for `Program Files (x86)/Steam`, `Program Files/Steam`, `Steam` on each drive.

`SteamNotFoundError`**:** Specific exception when the Steam installation cannot be located.

> **Note (F6-11):** Steam path discovery is duplicated in `ingestion/steam_locator.py` (primary). This module is supplementary (scans replay directories). Consolidation deferred; ensure same path precedence when modifying resolution.

-HLTV Module (`backend/data_sources/hltv/`)

The HLTV subsystem is composed of 5 specialized modules that collaborate to extract professional statistics from HLTV.org, bypassing Cloudflare anti-scraping protections:

> **Analogy:** The HLTV module is like a **well-organized spy team** that gathers information about the world's top players. The `stat_fetcher` is the field agent who knows where to find the data. The `docker_manager` prepares the armored vehicle (FlareSolverr) to pass through checkpoints (Cloudflare). The `flaresolverr_client` is the specialized driver. The `rate_limiter` is the timekeeper who ensures the team does not draw attention by moving too fast. The `selectors` are the map indicating exactly where to find each piece of information on the page.

`HLTVStatFetcher` (`stat_fetcher.py`) — Main scraping orchestrator:

Method	Description
`fetch_top_players()`	Scrape Top 50 players page → list of profile URLs
`fetch_and_save_player(url)`	Full player statistics fetch + DB save
`_fetch_player_stats(url)`	Deep-crawl: main page + sub-pages (clutch, multikill, career)
`_parse_overview(soup)`	Parse main statistics (rating, KPR, ADR, etc.)
`_parse_trait_sections(soup)`	Parse Firepower, Entrying, Utility sections
`_parse_clutches(soup)`	Parse 1v1/1v2/1v3 clutch wins
`_parse_multikills(soup)`	Parse 3K/4K/5K counts
`_parse_career(soup)`	Parse historical rating by year

Statistics extracted and saved in `ProPlayerStatCard`:

Category	Statistics
Core	`rating_2_0`, `kpr` (Kill/Round), `dpr` (Death/Round), `adr` (Damage/Round)
Efficiency	`kast` (Kill/Assist/Survival/Trade %), `headshot_pct`, `impact`
Opening	`opening_kill_ratio`, `opening_duel_win_pct`
Traits (JSON)	Firepower (kpr_win, adr_win), Entrying (traded_deaths_pct), Utility (flash_assists)
Insights (JSON)	Clutch (1on1/1on2/1on3), Multikill (3k/4k/5k), Career (rating per period)

`RateLimiter` (`rate_limit.py`) — 4-level rate limiting with anti-detection jitter:

Level	Min–Max Delay	Use case
micro	2.0s – 3.5s	Fast consecutive requests
standard	4.0s – 8.0s	Navigation between player profiles
heavy	10.0s – 20.0s	Transitions between sections (main → clutch → multikill → career)
backoff	45.0s – 90.0s	Suspected block or failure (graceful degradation)

> **Note (F6-25):** The jitter (`random.uniform(-0.5, 0.5)`) is **intentionally unseeded** — deterministic jitter would be detected by anti-scraping systems as an artificial pattern. The 2.0s minimum floor is always applied.

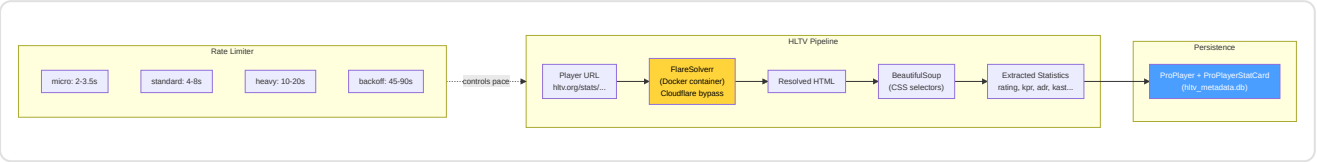
`DockerManager` (`docker_manager.py`) — FlareSolverr container management with cascading startup strategy:

1. **Fast path:** Returns `True` if already healthy (health check on `http://localhost:8191/`)
2. **Docker start:** Attempts `docker start flaresolverr` (15s timeout)

3. **Docker Compose fallback:** Attempts `docker-compose up -d` (60s timeout)
4. **Health polling:** Verifies availability every 3s for max 45s

`FlareSolverrClient` (`flaresolverr_client.py`) — Automatic bypass of Cloudflare JavaScript challenges. All HTTP requests are routed through FlareSolverr on `http://localhost:8191/` . The resolved HTML is passed to BeautifulSoup for parsing.

`selectors` (`selectors.py`) — CSS selectors for scraping HLTV pages, centralized for maintainability.



> **Architectural note:** The complete HLTV subsystem (with `HLTVApiService` , `CircuitBreaker` , `BrowserManager` , `CacheProxy` , `collectors`) resides in `ingestion/hltv/` and is documented in Part 3. The files in `data_sources/hltv/` are the low-level implementation of scraping and rate limiting.

> **HLTV database status (April 2026):** The `hltv_metadata.db` database contains **161 real professional players**, **32 teams**, and **156 stat cards** collected from live scraping of hltv.org via FlareSolverr. The CSS selectors in `selectors.py` are equipped with fallback chains to resist site layout changes. The `HybridCoachingEngine` uses these data for automatic reference pro selection: when generating an analysis, it automatically finds the pro player whose `rating_2_0` is closest to the user's and names them in the feedback ("your ADR is lower than [pro name]'s"), via `_find_best_match_pro()` in `coaching_service.py` and `_get_pro_name()` in `hybrid_engine.py` .

`hltv_scraper.py` / **`hltv_metadata.py`** (entry point in `data_sources/`):

- `run_hltv_sync_cycle(limit=20)` — Sync cycle orchestrator that imports `HLTVApiService` from the full pipeline
- `hltv_metadata.py` — Debug script for page saving via Playwright (CSS selector validation)

-FACEIT API and Integration (**`faceit_api.py`** , **`faceit_integration.py`**)

`faceit_api.py` **:** Single function `fetch_faceit_data(nickname)` that retrieves FACEIT Elo and Level for a given nickname. Requires `FACEIT_API_KEY` from configuration. Returns `{faceit_id, faceit_elo, faceit_level}` or empty dictionary on error.

`faceit_integration.py` **:** Complete FACEIT client with rate limiting:

Parameter	Value	Description
`BASE_URL`	`https://open.faceit.com/data/v4`	FACEIT v4 API endpoint
`RATE_LIMIT_DELAY`	6 seconds	10 req/min = 1 req every 6s (free tier)

`FACEITIntegration` class with:

- `_rate_limited_request(endpoint, params)` — requests with automatic rate limiting and exponential backoff on 429
- Match history management and demo download
- Dedicated exception `FACEITAPIError`

> **Analogy:** FACEIT is like an **external consultant** who provides the coach with a second opinion on the player's level. While the HLTV system provides data about professionals, FACEIT provides the competitive ranking of the user player (Elo and Level from 1 to 10). The rate limiting is like an **appointment with the consultant**: you cannot call more than 10 times per minute, otherwise the consultant refuses to respond (429 error). The system automatically respects this limit, waiting the necessary time between one request and the next.

-FrameBuffer — Circular Buffer for HUD Extraction

(`backend/processing/cv_framebuffer.py`)

The **FrameBuffer** is a thread-safe circular buffer for capturing and analyzing game screen frames. It functions as the "retina" of the system: it captures frames from the screen, stores them in a fixed-size ring, and allows the extraction of HUD (Head-Up Display) regions for visual analysis.

> **Analogy:** The FrameBuffer is like a **circular tape recorder** in a surveillance room. The camera (the game screen) records continuously, but the tape only has space for 30 frames — when it is full, new frames overwrite the older ones. The guard (the analysis system) can at any time ask "show me the last N frames" or "zoom into the minimap area in this frame". The important thing is that the recorder never blocks: even if the guard is analyzing a frame, the camera continues recording without interruption thanks to a lock that coordinates accesses.

Configuration:

Parameter	Default	Description
`resolution`	`(1920, 1080)`	Target frame resolution
`buffer_size`	`30`	Circular buffer capacity (frames)

Main operations:

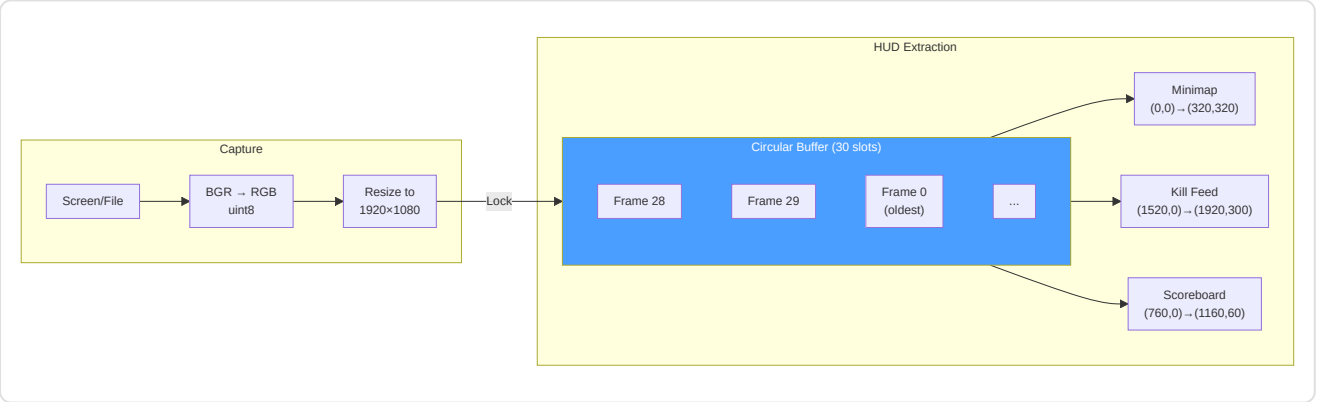
- `capture_frame(source)` — Ingests frame from file or numpy array → BGR→RGB, uint8, resize → push to circular buffer
- `get_latest(count=1)` — Retrieves the N most recent frames (newest to oldest)
- `extract_hud_elements(frame)` — Extracts all HUD regions into a dictionary

HUD Regions (1920×1080 reference):

Region	Coordinates	Position	Content
Minimap	`(0, 0, 320, 320)`	Top-left	CS2 radar (player positions)
Kill Feed	`(1520, 0, 1920, 300)`	Top-right	Kill feed and events
Scoreboard	`(760, 0, 1160, 60)`	Top-center	Team score

Resolution adaptation (`_scale_region()`): Coordinates are defined for the 1920×1080 reference resolution. For different resolutions, they are scaled proportionally: `sx = frame_width / 1920`, `sy = frame_height / 1080`. This makes the system **resolution-agnostic** — it works identically on 1080p, 1440p, or 4K monitors.

Thread-safety: A `threading.Lock()` protects all read and write operations on the buffer. The write index (`_write_index`) advances circularly modulo `buffer_size`, guaranteeing O(1) for insertion and retrieval.



-TensorFactory — Tensor Factory (`backend/processing/tensor_factory.py`)

The **TensorFactory** is the **perceptual system** of the RAP Coach: it converts raw game state into 3 image-tensors that the neural model can "see". Each tensor is a 3-channel image encoding a different dimension of the tactical situation: **map** (where everyone is), **view** (what the player can see), and **motion** (how they are moving).

> **Analogy:** The TensorFactory is like a **painter of military tactical maps** who receives radio reports and draws three separate maps for the commander (the RAP model). The first map (**tactical map**) shows the positions of known allies and enemies. The second map (**visibility map**) shows what the soldier can actually see from their point of view — the 90° cone in front of them. The third map (**motion map**) shows the soldier's recent path, their speed, and the direction of their crosshair. Crucially, the painter follows a strict rule: **they never draw the position of enemies the soldier has not seen** (NO-WALLHACK principle). If an enemy is behind a wall, they do not appear on the map — exactly as in the player's reality.

Configurations:

Parameter	`TensorConfig` (Inference)	`TrainingTensorConfig` (Training)
`map_resolution`	128 × 128	64 × 64
`view_resolution`	224 × 224	64 × 64
`sigma` (Gaussian blur)	3.0	3.0
`fov_degrees`	90°	90°
`view_distance`	2000.0 world units	2000.0 world units

> **Note (F2-02):** `TrainingTensorConfig` reduces resolution from 128/224 to 64/64, achieving a **memory saving of ~12×**. The `AdaptiveAvgPool2d` contract in RAPPerception produces 128-dim regardless of input resolution, but this guarantee is implicit — a runtime assertion is recommended.

Rasterization constants:

Constant	Value	Purpose
`OWN_POSITION_INTENSITY`	1.5	Brightness of own position marker
`ENTITY_TEAMMATE_DIMMING`	0.7	Teammates rendered darker than enemies
`ENTITY_MIN_INTENSITY`	0.2	Minimum intensity of visible entity
`ENEMY_MIN_INTENSITY`	0.3	Minimum intensity of visible enemy
`BOMB_MARKER_RADIUS`	50.0	Bomb circle radius (world units)
`BOMB_MARKER_INTENSITY`	0.8	Bomb circle opacity
`TRAJECTORY_WINDOW`	32 ticks	Trajectory window (~0.5s at 64 Hz)
`VELOCITY_FALLOFF_RADIUS`	20.0	Grid cells for radial velocity fade
`MAX_SPEED_UNITS_PER_TICK`	4.0	CS2 maximum speed (64 ticks/s)
`MAX_YAW_DELTA_DEG`	45.0	Flick threshold for aim detection

The 3 Rasterizers:

1. Map Rasterizer — `generate_map_tensor(ticks, map_name, knowledge)` → `Tensor(3, res, res)`

Channel	Player-POV Mode (with PlayerKnowledge)	Legacy Mode (no knowledge)
Ch0	Teammates (always known) + own position (intensity 1.5)	Enemy positions
Ch1	Visible enemies (full intensity) + last-known enemies (exponential decay)	Teammate positions
Ch2	Utility zones (smoke/molotov) + bomb overlay	Player position

2. View Rasterizer — `generate_view_tensor(ticks, map_name, knowledge)` → `Tensor(3, res, res)`

Channel	Player-POV Mode	Legacy Mode
Ch0	FOV mask (geometric 90° cone from gaze direction)	FOV mask
Ch1	Visible entities: teammates (dimmed ×0.7) + visible enemies (intensity weighted by distance)	Danger zone (areas NOT covered by accumulated FOV, capped at 8 ticks)
Ch2	Active utility zones (smoke/molotov circles in world units)	Safe zone (recently visible but not in current FOV)

3. Motion Encoder — `generate_motion_tensor(ticks, map_name)` → `Tensor(3, res, res)`

Channel	Content
Ch0	Trajectory of last 32 ticks — intensity ∝ recency (newest = 1.0, oldest → 0)
Ch1	Velocity field — radial gradient from player, modulated by current speed [0, 1]
Ch2	Crosshair movement — yaw delta magnitude as Gaussian blob at player position

> **Note (F2-03):** 128 tick/s demos compress velocity in the lower half of the [0, 1] range; tick-rate aware normalization pending implementation.

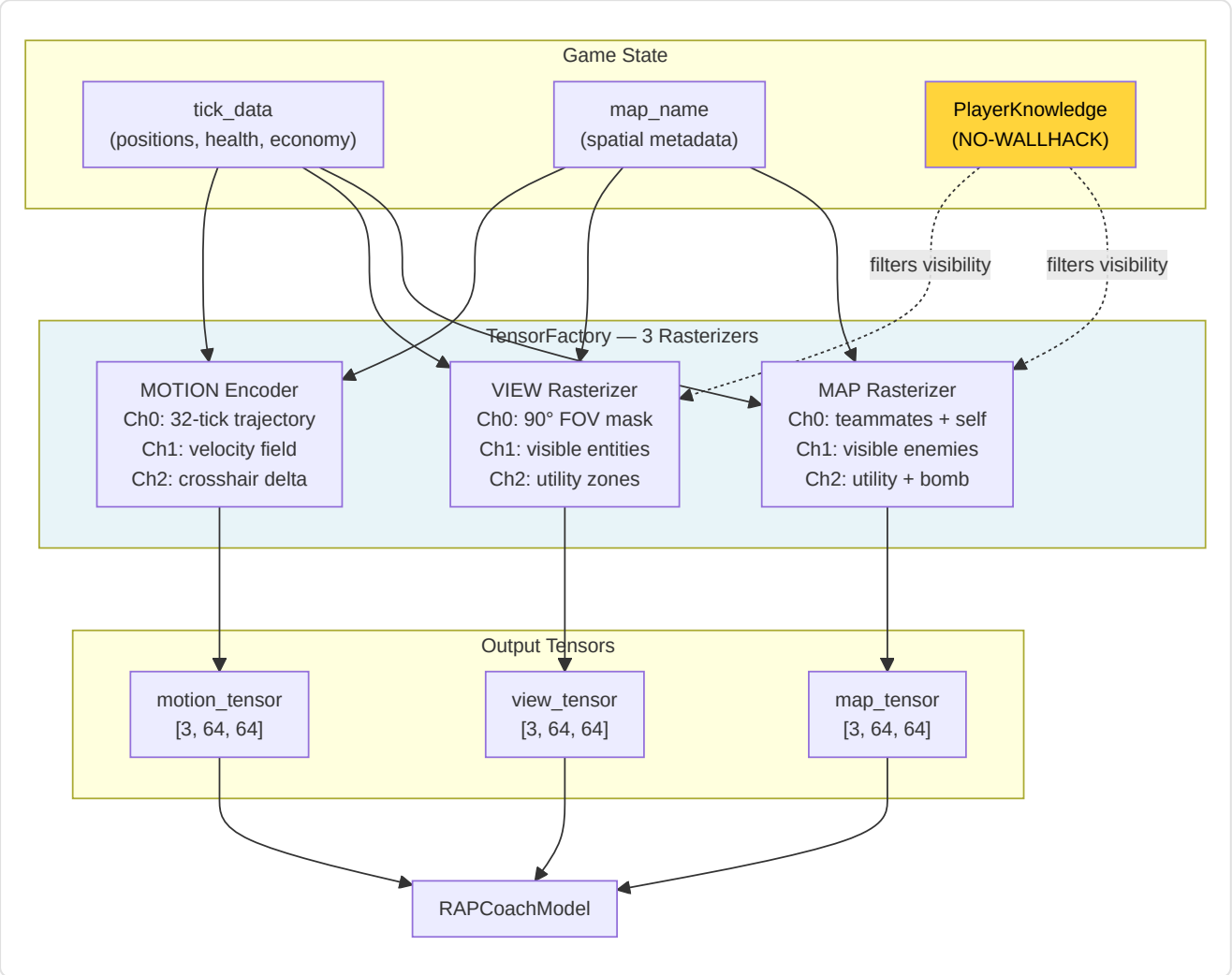
NO-WALLHACK integration: When `PlayerKnowledge` is provided, the map and view rasterizers encode **only the state visible to the player**. Last-seen enemy positions decay exponentially over time. Utility zones are visible only if in FOV or known from radar. When `knowledge=None`, the system degrades to legacy mode for backward compatibility.

Helper methods:

- `_world_to_grid(x, y, meta, resolution)` — World → grid coordinate conversion. **Note C-03:** Single Y-flip (`meta.pos_y - y`) to avoid double inversion
- `_normalize(arr)` — Normalization to [0, 1]. **Note M-10:** `arr / max(max_val, 1.0)` to prevent noise amplification in sparse channels

- `_generate_fov_mask(player_x, player_y, yaw, meta, resolution)` — 90° conical mask from gaze direction, distance-limited (top-down 2D approximation)

Singleton access: `get_tensor_factory()` — double-checked locking, thread-safe.



-FAISS Vector Index (`backend/knowledge/vector_index.py`)

The **VectorIndexManager** provides high-speed semantic search for the coach's RAG (Retrieval-Augmented Generation) knowledge system. It uses FAISS (Facebook AI Similarity Search) with `IndexFlatIP` on L2-normalized vectors, effectively achieving **cosine similarity search** in sub-linear time.

> **Analogy:** The FAISS index is like the coach's **library search system**. Instead of leafing through every book (tactical knowledge) or every note (coaching experience) one by one to find the one relevant to the current situation, the librarian (FAISS) has created an **index by concepts**: when the coach asks "what is the best strategy for a B retake on Mirage with 2 players?", the index instantly finds the 5 documents most similar to this question, without having to read all 10,000 documents in the library. The trick is that every document and every question is converted into a vector of 384 numbers (embedding), and FAISS compares these vectors via **inner product** (equivalent to cosine similarity after L2 normalization).

Dual indexes:

Index	DB Source	Content
`"knowledge"`	`TacticalKnowledge` table	Tactical knowledge embedding (strategies, positions, utility)
`"experience"`	`CoachingExperience` table	Coaching experience embedding (feedback, corrections, advice)

Index type: `faiss.IndexFlatIP` (Inner Product) on L2-normalized vectors. Since `cos(a, b) = a·b / (||a|| × ||b||)`, normalizing vectors to unit norm makes the inner product **exactly equivalent** to cosine similarity. Result range: [0, 1] where 1 = identical.

Public API:

Method	Description
`search(index_name, query_vec, k)`	Search the k most similar vectors. Lazy rebuild if dirty. Returns `List[(db_id, similarity)]`
`rebuild_from_db(index_name)`	Complete index rebuild from DB table. Thread-safe. Returns vector count
`mark_dirty(index_name)`	Marks the index for lazy rebuild (at next `search()`)
`index_size(index_name)`	Returns `index.ntotal` or 0 if not built

Disk persistence:

- Format: `{persist_dir}/{index_name}.faiss` + `{index_name}_ids.npy`
- Save: `faiss.write_index()` + `np.save()`
- Load: automatic in `__init__` via `faiss.read_index()` + `np.load()`
- Default directory: `~/.cs2analyzer/indexes/`

Thread-safety: A single `threading.Lock()` protects all read/write operations on the indexes, dirty flags, and rebuild operations. FAISS `IndexFlatIP` is thread-safe for concurrent reads.

Lazy rebuild (`mark_dirty()`): When new data is inserted into the Knowledge or Experience tables, the index is marked as "dirty" rather than rebuilt immediately. The rebuild happens only at the next `search()`, avoiding multiple rebuilds during batch insertions.

Vector normalization:

```
norms = ||embedding||₂ per row
normalized = embedding / max(norms, 1e-8)    # numerical stability
IndexFlatIP.add(normalized)
```

Graceful fallback: If `faiss-cpu` is not installed, the singleton `get_vector_index_manager()` returns `None` and the system automatically degrades to brute-force search (slower but functionally equivalent). This allows the program to work even on systems where FAISS is not available.

Over-fetching with explicit constants: To handle post-filtering scenarios (category, map_name, confidence, outcome), the search retrieves more results than necessary: `k × OVERFETCH_KNOWLEDGE = k × 10` for the Knowledge Base (filter by category + map), `k × OVERFETCH_EXPERIENCE = k × 20` for the Experience Bank (filter by map + confidence + outcome + composite scoring). The 20× multiplier for experiences is double that of knowledge because the filters are more restrictive (4 criteria vs 2), so a wider initial pool is needed to guarantee enough results after filtering.

-Round Context (`round_context.py`)

The **Round Context** module is the **temporal grid** of the ingestion system: it converts raw ticks from demo files into meaningful "round N, time T seconds" coordinates that every other module can use to contextualize game events.

> **Analogy:** Round Context is like the **timekeeper's assistant** in a soccer match. The timekeeper (DemoParser) measures time in absolute milliseconds from the start of the recording, but the assistant translates those milliseconds into useful information: "This event happened in the 23rd minute of the second half". Without the assistant, every analyst would have to make this conversion themselves, risking errors and inconsistencies. Round Context does the same for CS2: it converts absolute ticks into "Round 7, 42 seconds from the start of the action", allowing all analysis engines to work with consistent and meaningful temporal coordinates.

Public functions:

Function	Input	Output	Complexity
`extract_round_context(demo_path)`	`.dem` file path	DataFrame: `round_number`, `round_start_tick`, `round_end_tick`	O(n) event parsing
`extract_bomb_events(demo_path)`	`.dem` file path	DataFrame: `tick`, `event_type` (planted/defused/exploded)	O(n) event parsing
`assign_round_to_ticks(df_ticks, round_context, tick_rate)`	Tick DataFrame + round boundaries	DataFrame enriched with `round_number`, `time_in_round`	O(n log m) via `merge_asof`

Round boundary construction (`extract_round_context`):

The module analyzes two types of events from the demo file:

- `round_freeze_end` — the tick at which freeze time ends and the action starts (players can move)
- `round_end` — the tick at which the round ends (win/loss)

For each round, it pairs the last `round_freeze_end` preceding the corresponding `round_end`. **Fallback:** if no `round_freeze_end` event is found for a given round (possible in corrupted demos or interrupted matches), it uses the previous round's `round_end` as start, logging a warning.

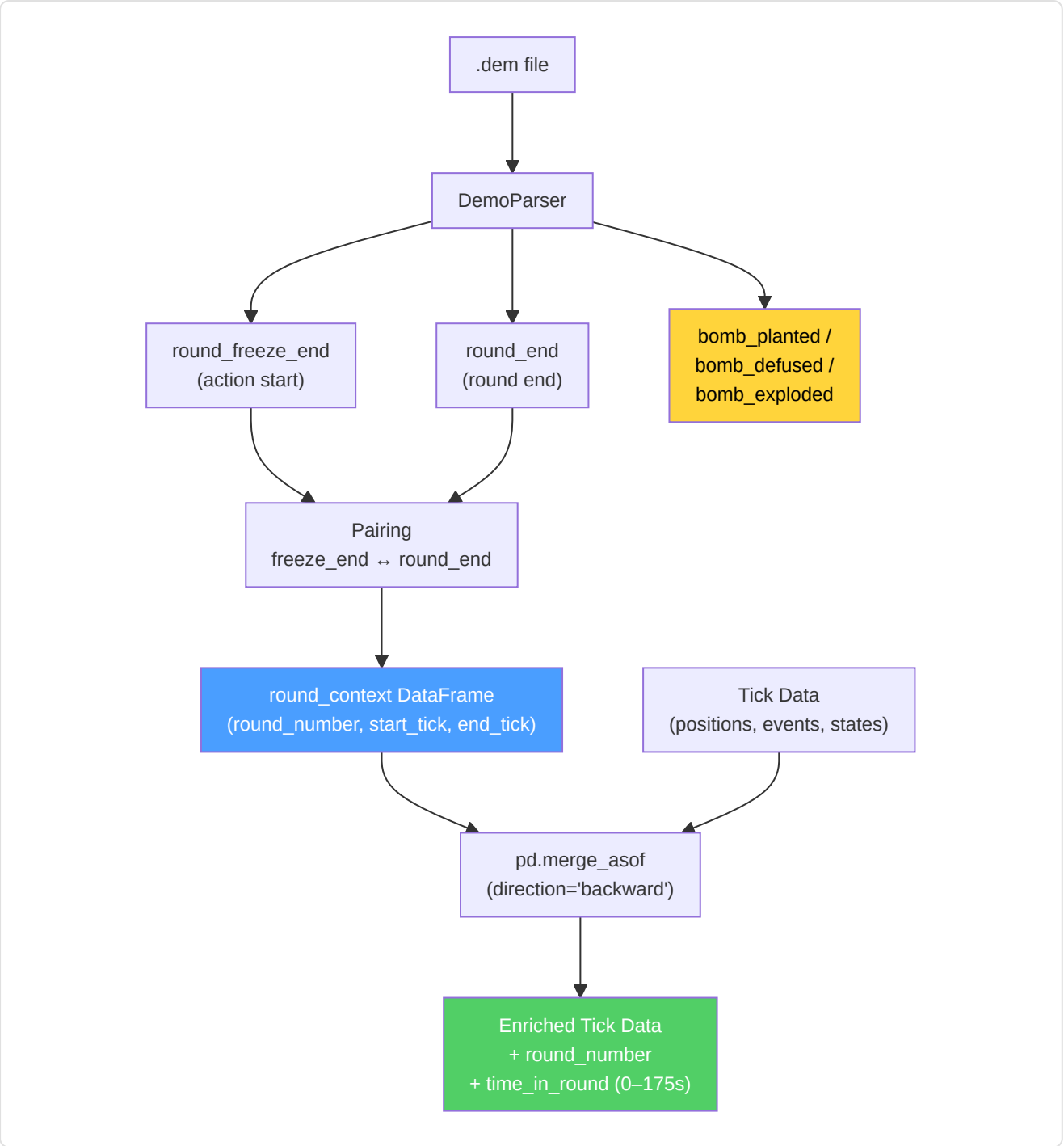
Bomb event extraction (`extract_bomb_events`):

Extracts three types of events: `bomb_planted`, `bomb_defused`, and `bomb_exploded`. The addition of `bomb_exploded` (H-07 remediation) makes it possible to distinguish between rounds won by explosion and rounds won by elimination, information critical for post-plant tactical analysis.

Round assignment to ticks (`assign_round_to_ticks`):

Uses `pd.merge_asof` with `direction="backward"` for efficient O(n log m) assignment: for each tick, find the last `round_start_tick ≤ tick`. Computes `time_in_round = (tick − round_start_tick) / tick_rate`, clamped to [0.0, 175.0] seconds (maximum duration of a CS2 round). Ticks before the first round (warmup) are assigned to round 1.

> **Note:** Using `merge_asof` instead of a Python loop transforms an O(n × m) operation into O(n log m), fundamental for demos with millions of ticks and 30+ rounds.



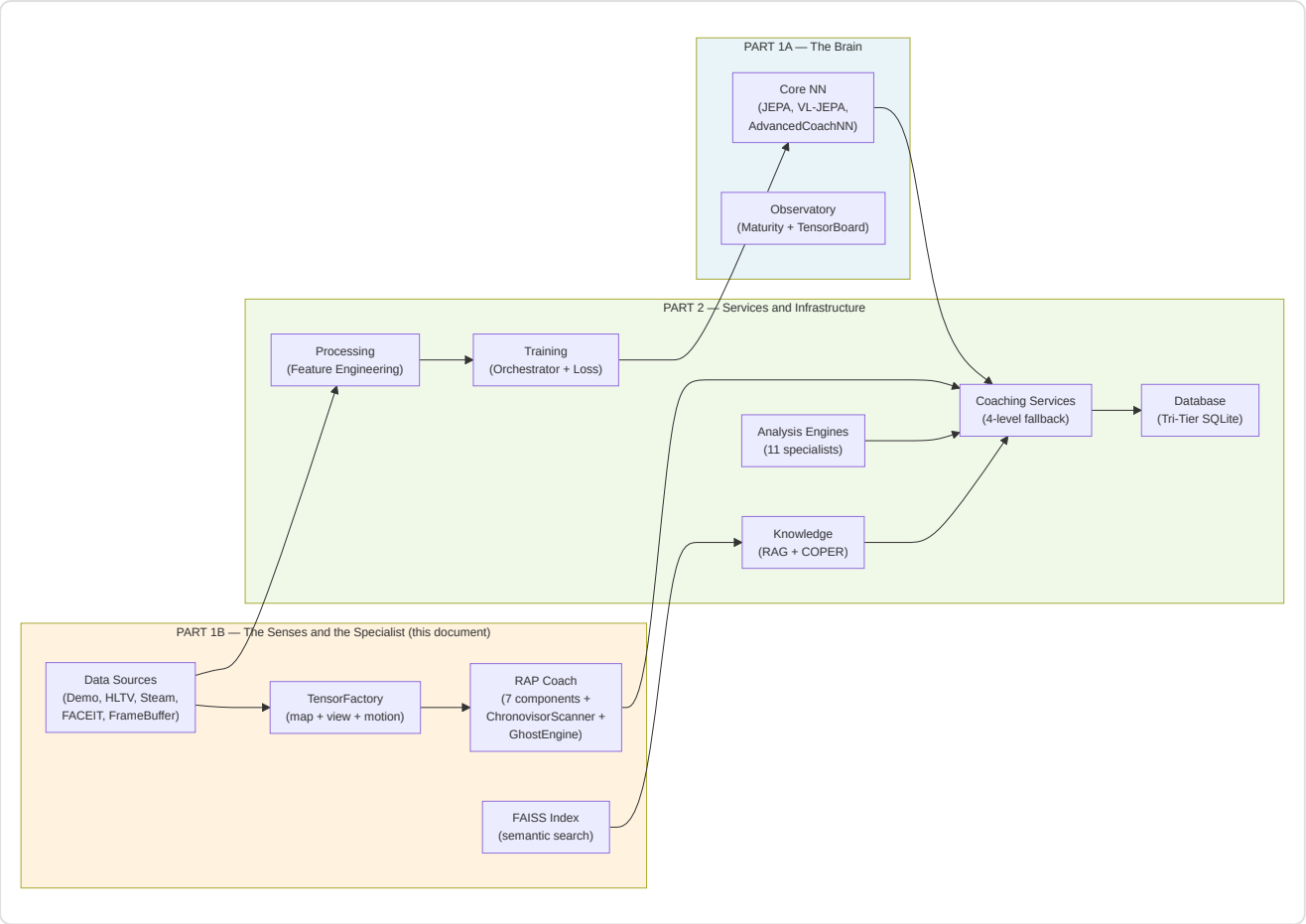
Error handling: Each parsing phase is protected by try/except with structured logging. If parsing fails completely or no `round_end` events are found, the function returns an empty DataFrame — downstream modules (e.g., `RoundStatsBuilder`) must handle this case gracefully.

Summary of Part 1B — The Senses and the Specialist

Part 1B has documented the **two perceptual and diagnostic pillars** of the coaching system:

Subsystem	Role	Key Components
2. RAP Coach	The **specialist doctor** — 7-component architecture for complete coaching under POMDP conditions	Perception (3-stream ResNet, 24 conv), Memory (LTC **512** NCP units + Hopfield 4 heads + NN-MEM-01 activation delay + **RAPMemoryLite** LSTM fallback), Strategy (4 MoE experts + SuperpositionLayer), Pedagogy (Value Critic + Skill Adapter), Causal Attribution (5 categories, learned utility signal), Positioning (Linear 256→3), Communication (template), ChronovisorScanner (3 temporal scales + 50K tick safety limit + cross-scale deduplication + structured ScanResult), GhostEngine (4-tensor pipeline with POV mode R4-04-01, hidden_state NN-40, 5-level fallback)
1B. Data Sources	The **senses** — acquire and structure data from the outside world	Demo Parser (demoparser2 + HLTV 2.0 rating), Demo Format Adapter (PBDEMS2 magic bytes), Event Registry (complete CS2 schema), Trade Kill Detector (192-tick window), Steam API (retry + backoff), Steam Demo Finder (cross-platform), HLTV (FlareSolverr + 4-level rate limiting + CSS selectors with fallback chain — **161 real pro players, 32 teams, 156 stat cards** in hltv_metadata.db), FACEIT API, FrameBuffer (30-frame ring buffer), TensorFactory (3 NO-WALLHACK rasterizers), FAISS (IndexFlatIP 384-dim), Round Context (merge_asof O(n log m))

Final analogy: If the coaching system were a **human being**, Part 1A described its brain (the neural networks that learn and the maturity system that decides when they are ready), and Part 1B has described its eyes and ears (the data sources that acquire information from the outside world), its specialized nervous system (the RAP Coach that integrates perception, memory, and decision), and its communication system (which translates understanding into readable advice). But a brain with senses alone is not enough: it needs a **body** to act. **Part 2** documents that body — the services that synthesize advice, the analysis engines that investigate every aspect of gameplay, the knowledge systems that store accumulated wisdom, the processing pipeline that prepares the data, the database that preserves everything, and the training pipeline that teaches the models.



Continues in Part 2 — *Coaching Services, Coaching Engines, Knowledge and Retrieval, Analysis Engines (11), Processing and Feature Engineering, Control Module, Progress and Trends, Database and Storage (Tri-Tier), Training and Orchestration Pipeline, Loss Functions*